

LATTICE GAUGE THEORY: RENORMALIZATION TO ALL ORDERS IN THE LOOP EXPANSION

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The renormalization of $SU(N)$ lattice gauge theories is discussed. The program is based on lattice power counting theorems. By choice of an admissible gauge fixing, e.g. the Lorentz gauge, the theory exhibits a BRS-symmetry. It is proven that the existence of such a symmetry insures finiteness of the continuum limit by simply renormalizing the coupling constant and field strengths, to every order in perturbation theory.

1. Introduction

Lattice gauge theories are gauge theories with a non-perturbative, gauge invariant ultraviolet (UV) cutoff. That allows one to treat gauge theories in a non-perturbative way. Observable physics has to be searched for in the limit of vanishing lattice spacing a . Due to the complexity and difficulty of that limit one is often forced to use conventional perturbative methods. Actually perturbation theory is expected to be reliable for reasonable large cutoff $\Lambda = a^{-1}$ [1], in a region of momentum space which depends on the ultraviolet and infrared (IR) behavior of the theory, e.g. a non-abelian gauge theory is anticipated to be UV-asymptotically free.

In the limit $\Lambda \rightarrow \infty$ one has to deal with renormalization in order to make the limit theory finite. An artifact of the lattice cutoff is the specific structure of Feynman integrals which prevent the conventional power counting theorems [2–7] to apply. Nevertheless, we have appropriate theorems at our disposal [8–11] which allow power counting also for lattice diagrams. That includes the behavior of “irrelevant” Feynman integrals, i.e. those which contain vertices resulting from contributions to the lattice action with vanishing continuum limit. Such diagrams, too, must be renormalized. Their number grows with increasing number of loops. The ideas of renormalization apply also to non-polynomial local interactions such as

lattice gauge theory. To every finite order of the Planck constant $\hbar$ the theory is described by a polynomial interaction which we get by expanding the action in powers of small gauge fields (to an order depending on the power of $\hbar$). In general, the constraints on the action for power counting renormalizability are that it has a local continuum limit and that the higher order terms in the lattice spacing a do not invalidate the leading order power counting. That means every additional power of the fields or lattice momenta should be accompanied by a corresponding power of a [9, 11]. The renormalized continuum field theory is then determined by the continuum limit of the renormalized lattice theory, to every order in perturbation theory.

Typically, in order to apply perturbation theory, a gauge theory needs gauge fixing which divides out the unphysical gauge degrees of freedom. We employ only linear gauge fixing functions, in particular the Lorentz gauge, which transform under the adjoint representation of the global gauge group, so that the corresponding gauge fixed actions are invariant under such transformations. This way we avoid the technical complications arising by other gauges (gauge field and Faddeev–Popov mass, four-ghost vertices etc. [12–14]). The Faddeev–Popov trick introduces ghost degrees of freedom due to the non-abelian nature of the transformation laws for $SU(N)$ gauge fields we are concerned with ($N \geq 2$). Under the above-mentioned constraints the theory is renormalizable by power counting, i.e. with increasing number of loops the order of subtractions needed does not increase. They can always be chosen in such a way that the renormalized theory is IR- and UV-convergent [11], in the sum of all diagrams to a given order. However, a priori only little can be said about the structure of the counterterms. If compared to “continuum regularizations” the vertices are of a more complicated form. There are many more diagrams due to irrelevant vertices. Their number grows with increasing number of loops. Also, we have to deal with contributions from the gauge invariant functional measure, which cannot be neglected. In particular, these terms are important to insure that the self-energy of the gluon vanishes at zero momentum, i.e. the absence of a mass term, as must be for IR-convergence.

Nevertheless, all these complications are not completely arbitrary. Clearly, as far as we are concerned only with the continuum limit itself we are not forced to use a specific gauge invariant lattice action, for instance an action of the Wilson type. We are also allowed to use e.g. actions which include traces around other than simple plaquettes. The only a priori condition is that in the large cutoff limit the action should reproduce the continuum gauge field action. On the other hand, we should expect that for any gauge invariant cutoff the introduction of a gauge fixing followed by the Faddeev–Popov trick leads to a BRS-type symmetry [15, 16]. That is simply a consequence of the gauge fixing procedure in presence of a gauge invariant cutoff. Such a symmetry manifests itself in relations between basic field Green functions and those containing composite operator insertions which describe the non-linear symmetry transformations of the fields. They are usually combined into differential equations for the generating functionals. Many features of the theory

derive from these Ward identities (WI's). In particular, all afore-mentioned Green functions are IR-convergent, as a consequence of the fact that two- and three-point functions vanish at zero momentum.

The idea we follow here is to use the WI's in order to determine the structure of the counterterms needed to render the theory finite. The procedure is by induction on the number of loops. At a given order, taking into account all the counterterms of lower order, the "overall divergencies" are roughly determined by power counting, and their specific forms derive from the WI's. Our aim is to show that all the counterterms can be chosen in such a way that the theory is renormalized simply by adjusting the parameters of the bare action. Clearly that would allow for many additional renormalizations, appropriately adjusted. In fact, as we shall see below, this is possible independently of the originally chosen gauge field action (under the general constraints mentioned above), taking into account universality of the continuum limit. Furthermore, the renormalized theory will satisfy similar symmetry relations as the bare one, which implies IR-finiteness. In particular, the transformation laws are multiplicatively renormalized.

The basic ingredient used throughout is the presence of a symmetric UV-cutoff [12,17]. It allows us to derive the structure of the divergencies. Also, other basic tools as equations of motion or variation principles are easily derived.

The article is organized as follows. In sect. 2 we introduce general notations currently used. Gauge fixing is introduced, and we give the partition function of pure gauge theory on a four-dimensional hypercubic lattice in the infinite volume we are concerned with in terms of the gauge fixed effective action. It exhibits a BRS-symmetry which for the gauge field corresponds to an infinitesimal gauge transformation. In sect. 3 the WI's are derived and the equations of motion for the Faddeev–Popov field are given which are of use later on. In sect. 4 we state that the theory can be renormalized simply by adjusting the parameters of the bare action (coupling constant and field strengths) as appropriate functions of the cutoff and renormalized parameters. There are only three independent renormalization constants usually to be fixed by appropriate normalization conditions. Actually, the gauge field measure as it stands prevents the power counting theorems to apply directly. As shown in sect. 5 that obstacle is removed by introducing massive auxiliary Grassman fields which do not propagate and have IR- and UV-dimension two. That means they produce no IR-singularities. The lattice power counting theorems now apply and we state an auxiliary theory which describes the renormalization of the action including the additional ghosts. That statement is proved in sect. 6. The renormalization theorem then follows immediately by doing the gaussian integration on the auxiliary fields. In sect. 7 we check that the conventional result that the β -function is independent of the gauge fixing in the minimal renormalization scheme holds also for the lattice cutoff. Finally, it is pointed out in sect. 8 that lattice fermions do not invalidate renormalizability of gauge theory as long as the propagators have only one pole in the Brillouin zone. That can be

achieved e.g. by Wilson's projection operator technique. The conclusions are drawn in sect. 9.

2. Gauge fixing and BRS-symmetry

We first introduce some general notations. Let us consider pure $SU(N)$ gauge theory, $N \geq 2$, on an infinite four-dimensional hypercubic lattice aZ^4 , where a denotes the lattice spacing in some externally defined physical units. Let

$$T_j, \quad j = 1, \dots, N^2 - 1 \quad (2.1)$$

denote a basis of $\mathfrak{su}(N)$, the Lie algebra of $SU(N)$. We assume normalization

$$\text{tr}(T_a T_e) = -\delta_{ae}. \quad (2.2)$$

To every link (x, μ) , where $x \in aZ^4$ and $\mu \in \{1, \dots, 4\}$, we associate an element

$$\begin{array}{ccc} \bullet & \longleftarrow & \bullet \\ x & & x + a\hat{\mu} \end{array} = U(x, \mu) \in SU(N) \quad (2.3)$$

The symmetry group

$$G = \bigotimes_x SU(N)$$

of gauge transformations

$$\Lambda(x) \in SU(N)$$

acts on such fields by

$$U(x, \mu) \rightarrow {}^A U(x, \mu) = \Lambda(x) U(x, \mu) \Lambda^{-1}(x + a\hat{\mu}), \quad (2.4)$$

where $\hat{\mu}$ is a unit-vector in μ -direction. The Lie algebra $\mathcal{G}$ of G consists of the fields

$$\omega(x) \in \mathfrak{su}(N)$$

for all $x \in aZ^4$. In $\mathcal{G}$ we define the symmetric scalar product

$$(X, Y)_{\mathcal{G}} = -a^4 \sum_x \text{tr}(X(x) Y(x)); \quad X, Y \in \mathcal{G}. \quad (2.5)$$

Lattice derivatives are defined by

$$\begin{aligned} \partial_{\mu} f(x) &= f(x + a\hat{\mu}) - f(x), \\ \partial_{\mu}^* f(x) &= f(x) - f(x - a\hat{\mu}). \end{aligned} \quad (2.6)$$

Notice that

$$\partial_\mu(fg)(x) = (\partial_\mu f(x))g(x) + f(x)(\partial_\mu g(x)) + (\partial_\mu f(x))(\partial_\mu g(x)).$$

We are concerned with perturbation theory on infinite volume lattice gauge theory. The following calculations may be thought of as in finite volume, the limit of infinite volume taken at the end. Let us consider the expectation value of an arbitrary gauge invariant quantity $I(U)$,

$$\langle I \rangle = [I]/[1], \quad (2.7)$$

$$[I] = \int \mathcal{D}U I(U) \exp\{-S(U)\},$$

$$[1] = \int \mathcal{D}U \exp\{-S(U)\},$$

where

$$\mathcal{D}U = \prod_{x, \mu} dU(x, \mu), \quad (2.8)$$

and $dU(x, \mu)$ denotes the invariant measure on $SU(N)$. The action $S(U)$ is assumed to be invariant under gauge transformations (2.4) and continuous in the group elements U . For instance, S may be the Wilson action. We are concerned with conventional perturbation theory, i.e. with a saddle point expansion about the classical vacuum. We thus restrict attention to an (infinitesimally) small neighborhood Ω of the pure gauge configurations

$$\Lambda(x)\Lambda^{-1}(x + a\hat{\mu}). \quad (2.9)$$

The gauge field $\mathcal{A}$ appears as a Lie algebra valued vector field

$$\mathcal{A}_\mu(x) = \sum_d \mathcal{A}_\mu^d(x) T_d \quad (2.10)$$

as a local parametrization

$$U(x, \mu) = \exp(ag\mathcal{A}_\mu(x)). \quad (2.11)$$

g is the (bare) coupling constant. Under an infinitesimal gauge transformation, the parametrization changes according to

$$e^{ag(\mathcal{A}_\mu(x) + \delta_{(\omega)}\mathcal{A}_\mu(x) + O(\omega^2))} = e^{\omega(x)} e^{ag\mathcal{A}_\mu(x)} e^{-\omega(x + a\hat{\mu})}, \quad (2.12)$$

thus the gauge field $\mathcal{A}$ transforms as

$$\delta_{(\omega)} \mathcal{A}_\mu(x) = -\frac{1}{a} \left[\text{ad}(\text{ag} \mathcal{A}_\mu(x)) + \frac{\text{ad}(\text{ag} \mathcal{A}_\mu(x))}{1 - \exp\{-\text{ad}(\text{ag} \mathcal{A}_\mu(x))\}} \partial_\mu \right] \omega(x), \quad (2.13)$$

where $\text{ad } X(Y) = [X, Y]$ for $X, Y \in \text{su}(N)$. Notice that

$$[\delta_{(\omega)}, \delta_{(\nu)}] h(\mathcal{A}) = -g \delta_{([\omega, \nu])} h(\mathcal{A}) \quad (2.14)$$

for any smooth function h of $\mathcal{A}$, as a consequence of the group property of gauge transformations.

We shall give only a rough outline of the gauge fixing procedure in the perturbative framework and of the derivation of the Faddeev–Popov action. A more detailed derivation can be found in refs. [18, 29]. In order to separate the (unphysical) gauge degrees of freedom from the physical ones we must introduce an appropriate gauge. Let $F: \Omega \rightarrow \mathcal{G}$ be an admissible gauge fixing function, i.e. for almost all sufficiently small $X \in \mathcal{G}$ and almost all $U \in \Omega$ there is one and only one $\Lambda \in G$, so that

$$F({}^\Lambda U) = X.$$

In infinite volume and for small enough Ω the Lorentz gauge

$$F(U) = (1/a) \sum_\mu \partial_\mu^* \mathcal{A}_\mu(x) \quad (2.15)$$

is such an admissible gauge. For, the neighborhood Ω of the gauge orbit admits a parametrization

$$U(x, \mu) = \Lambda(x) e^{\text{ag} \mathcal{A}_\mu(x)} \Lambda^{-1}(x + a\hat{\mu}),$$

where $\Lambda(x) \in \text{SU}(N)$ and

$$(1/a) \sum_\mu \partial_\mu^* \mathcal{A}_\mu(x) = X(x),$$

$X \in \mathcal{G}$ a given small field, and we impose the constraint “ $\Lambda(x) \rightarrow 1$ as $x \rightarrow \infty$ ”.

Now let $X \in \mathcal{G}$ be arbitrary. We insert in the numerator and denominator of (2.7) the constant

$$\int \mathcal{D}\Lambda \delta(F({}^\Lambda U) - X) \det L({}^\Lambda U),$$

where $L(U)$ is the jacobian of $F(U)$ under infinitesimal gauge transformations, i.e. $\delta_{(\omega)} F(U) = L(U)\omega$, and $\mathcal{D}\Lambda = \prod_x d\Lambda(x)$, $d\Lambda(x)$ being the invariant measure on

$SU(N)$. Using gauge invariance we get

$$[I] \sim \int \mathcal{D}U I(U) e^{-S(U)} \delta(F(U) - X) \det L(U).$$

By definition, $[I]$ is independent of X . We can thus take the mean over X with a gaussian weight

$$e^{-(\lambda/2)(X, X)_{\mathcal{G}}}, \quad \lambda > 0.$$

Using the parametrization (2.11), the group measure may be represented by [18, 19]

$$\mathcal{D}\mathcal{A}(\det G)^{1/2},$$

where

$$\mathcal{D}\mathcal{A} = \prod_{x, d, \mu} d\mathcal{A}_{\mu}^d(x), \quad G_{x\mu d, yve} = \left\langle U^{-1} \frac{\partial}{\partial \mathcal{A}_{\mu}^d(x)} U, U^{-1} \frac{\partial}{\partial \mathcal{A}_v^e(y)} U \right\rangle,$$

and for Lie algebra valued vector fields A, B ,

$$\langle A, B \rangle = -\text{tr} \sum_{x, \mu} A_{\mu}(x) B_{\mu}(x).$$

We thus get up to exponentially small corrections in g or $\hbar$ (cf eqs. (2.12), (2.13))

$$\langle I \rangle = (1/Z) \int \mathcal{D}\mathcal{A} \det J \cdot I \cdot e^{-S - (\lambda/2)(F, F)_{\mathcal{G}}} \cdot \det L, \quad (2.16)$$

where

$$Z = \int \mathcal{D}\mathcal{A} \det J e^{-S - (\lambda/2)(F, F)_{\mathcal{G}}} \cdot \det L \quad (2.17)$$

and

$$\det J = \exp \left\{ \sum_{x, \mu} \text{tr} \log \frac{1 - e^{-\text{ad}(ag\mathcal{A}_{\mu}(x))}}{\text{ad}(ag\mathcal{A}_{\mu}(x))} \right\}. \quad (2.18a)$$

The integration range of the gauge field $\mathcal{A}_{\mu}^d(x)$ now extends to infinity. Notice that

$$\mathcal{D}\mathcal{A} \det J \quad (2.18b)$$

is invariant under infinitesimal gauge transformations (2.13). Finally, the

Faddeev–Popov determinant $\det L$ is written as

$$\det L \cong \int \mathcal{D}c \mathcal{D}\bar{c} e^{-S_{\text{FP}}},$$

$$\mathcal{D}c \mathcal{D}\bar{c} = \prod_{x,d} dc_d(x) \cdot \prod_{x,d} d\bar{c}_d(x), \quad (2.19)$$

where $c_d(x), \bar{c}_d(x), x \in a\mathbb{Z}^4, d = 1, \dots, N^2 - 1$, are Grassman variables and in terms of $c(x) = \sum_d c_d(x) T_d, \bar{c}(x) = \sum_d \bar{c}_d(x) T_d$ we get e.g. in the Lorentz gauge

$$S_{\text{FP}} = (\bar{c}, \Delta_{\text{FP}}(\mathcal{A}; g) c)_{\mathcal{G}}$$

$$= a^4 \sum_{x,d,\mu} \bar{c}_d(x) \left(-\frac{1}{a^2} \right) \partial_\mu^* \left(\left[\text{ad}(ag\mathcal{A}_\mu(x)) \right. \right.$$

$$\left. \left. + \frac{\text{ad}(ag\mathcal{A}_\mu(x))}{1 - \exp\{-\text{ad}(ag\mathcal{A}_\mu(x))\}} \partial_\mu \right] c(x) \right)_d. \quad (2.20)$$

By the introduction of a gauge fixing gauge invariance has been lost. Nevertheless one should expect that the gauge fixed action exhibits a BRS-symmetry [15, 16]. That will be important for stability of the action under renormalization. For the lattice cutoff such a symmetry transformation is given by

$$\delta_\epsilon \mathcal{A}_\mu(x) = \epsilon \left(-\frac{1}{a} \right) \left(\text{ad}(ag\mathcal{A}_\mu(x)) + \frac{\text{ad}(ag\mathcal{A}_\mu(x))}{1 - \exp\{-\text{ad}(ag\mathcal{A}_\mu(x))\}} \partial_\mu \right) c(x)$$

$$\delta_\epsilon c(x) = \epsilon g c(x)^2, \quad \delta_\epsilon \bar{c}(x) = -\lambda \epsilon F(\mathcal{A}(x)). \quad (2.21)$$

ϵ is a Grassman variable, anti-commuting with the ghosts $c, \bar{c}$. In fact, $\delta_\epsilon \mathcal{A}_\mu$ is an infinitesimal gauge transformation, i.e. $\delta_\epsilon \mathcal{A}_\mu = \delta_{(\epsilon c)} \mathcal{A}_\mu$. Furthermore, using $\delta_\epsilon F = \epsilon \Delta_{\text{FP}} c$, we get

$$(\lambda \delta_\epsilon F, F)_{\mathcal{G}} + (\delta_\epsilon \bar{c}, \Delta_{\text{FP}} c)_{\mathcal{G}} = 0 \quad (2.22)$$

and

$$\delta_\epsilon (\Delta_{\text{FP}} c) = \delta_{(\epsilon c)} \Delta_{\text{FP}} c + \Delta_{\text{FP}} (\delta_\epsilon c)$$

$$= \delta_{(\epsilon c)} \delta_{(c)} F + g \epsilon \Delta_{\text{FP}} (c^2) = 0, \quad (2.23)$$

because of

$$\begin{aligned}\delta_{(\varepsilon c)}\delta_{(c)}F &= \tfrac{1}{2}[\delta_{(\varepsilon c)}, \delta_{(c)}]F = -(g/2)\delta_{[\varepsilon c, c]}F \\ &= -(g/2)\Delta_{\text{FP}}[\varepsilon c, c] = -g\varepsilon\Delta_{\text{FP}}(c^2).\end{aligned}$$

Hence the gauge fixed action is BRS-invariant. It will be shown below that this symmetry is stable under renormalization, that is, there is a renormalized lattice action which has such a symmetry. That will imply renormalizability of the theory by simply renormalizing the parameters of the bare action. Actually, a gauge fixed action derived à la Faddeev–Popov always exhibits invariance under the BRS-transformation (2.21) independent of which gauge fixing F has been chosen [16]. In particular, F needs not to be linear. On the other hand, renormalization is more involved if the gauge is not linear or the action lacks invariance under global gauge transformations [12–14]. In that case it cannot be expected at all that the theory is made finite simply by choosing the bare parameters as appropriate functions of the cutoff and the renormalized ones. To avoid such sophisticated complications we always assume an admissible gauge fixing F which is linear and transforms under the adjoint action of the global gauge group. Also, F should respect the discrete lattice symmetries which form a subgroup of the euclidean group of space–time transformations (translations, rotations and reflections). In particular, we employ the Lorentz gauge which satisfies all the criteria.

It is easy to show that the BRS-variations have the following nilpotence properties:

$$\delta_\varepsilon\delta_\eta\mathcal{A}_\mu = \delta_\varepsilon\delta_\eta c = 0, \quad (2.24)$$

where ε and η are Grassmann variables. For instance,

$$\delta_\varepsilon\delta_\eta\mathcal{A}_\mu = \delta_\varepsilon(\eta\Delta_{\text{FP}}c) = 0,$$

as shown in eq. (2.23). That implies the composite operators which describe the transformation laws of the basic fields are BRS-invariant. Generally, in order to derive WI's such operators must be included into the action, coupled to new sources, and the same holds for their variations under the transformation and so on, until the system closes [12,20]. Fortunately, the identities (2.24) imply that this procedure ends after the first step. Thus only two operators are to be included (the BRS-variation of the anti-ghost $\bar{c}$ is linear and does not need a new source, as long as the gauge fixing function is linear). As we shall see the WI is a non-linear first order differential equation for the vertex functional. That implies renormalization also of the transformation laws.

3. Ward identities

The previous considerations instruct us to set up perturbation theory by the following bare partition function with sources coupled to the basic fields and their transformations:

$$\begin{aligned} Z_0(J, \bar{\xi}, \xi, K, L; g, \lambda) \\ = N_0 \int \mathcal{D}\mathcal{A} \mathcal{D}c \mathcal{D}\bar{c} \exp \{ \mathcal{E}_0(\mathcal{A}, c, \bar{c}, K, L; g, \lambda) + (1/\hbar) S_c(\mathcal{A}, c, \bar{c}; J, \bar{\xi}, \xi) \} \end{aligned} \quad (3.1)$$

N_0 is the normalization

$$Z_0(\text{all sources} = 0; g, \lambda) = 1, \quad (3.2)$$

and the product measure is given by

$$\mathcal{D}\mathcal{A} \mathcal{D}c \mathcal{D}\bar{c} = \prod_{x, d, \mu} d\mathcal{A}_\mu^d(x) \cdot \prod_{x, e} dc_e(x) \cdot \prod_{x, e} d\bar{c}_e(x). \quad (3.3)$$

S_c is the source term for the fields,

$$S_c(\mathcal{A}, c, \bar{c}; J, \bar{\xi}, \xi) = a^4 \sum_{x, d} \left\{ \sum_\mu J_\mu^d(x) \mathcal{A}_\mu^d(x) + \bar{\xi}_d(x) c_d(x) + \bar{c}_d(x) \xi_d(x) \right\}. \quad (3.4)$$

Furthermore

$$\begin{aligned} \mathcal{E}_0(\mathcal{A}, c, \bar{c}, K, L; g, \lambda) = \sum_{x, \mu} \text{tr} \log \frac{1 - \exp[-\text{ad}(ag\mathcal{A}_\mu(x))]}{\text{ad}(ag\mathcal{A}_\mu(x))} \\ + (1/\hbar) a^4 \sum_{x, d} \left\{ \sum_\mu K_\mu^d(x) s\mathcal{A}_\mu^d(x) + L_d(x) sc_d(x) \right\} \\ - (1/\hbar) S_0(\mathcal{A}, c, \bar{c}; g, \lambda). \end{aligned} \quad (3.5)$$

Here we have included the contribution from the measure (2.18a) into the action $\mathcal{E}_0$. For a functional G of the fields the notation sG is defined by [13]

$$\delta_\epsilon G = \epsilon sG. \quad (3.6)$$

In particular,

$$s\mathcal{A}_\mu(x) = -\frac{1}{a} \left[\text{ad}(ag\mathcal{A}_\mu(x)) + \frac{\text{ad}(ag\mathcal{A}_\mu(x))}{1 - \exp[-\text{ad}(ag\mathcal{A}_\mu(x))]} \partial_\mu \right] c(x)$$

$$sc(x) = g(c(x))^2, \quad s\bar{c}(x) = -\lambda F(\mathcal{A}(x)). \quad (3.7)$$

Note that $s^2\mathcal{A}_\mu(x) = s^2c(x) = 0$, as follows from eq. (2.24). The bare action S_0 is given by

$$S_0(\mathcal{A}, c, \bar{c}; g, \lambda) = S_W(\mathcal{A}; g) + S_{\text{gf}}(\mathcal{A}; \lambda) + S_{\text{FP}}(\mathcal{A}, c, \bar{c}; g), \quad (3.8)$$

where S_W denotes the gauge field action which is a continuous function of the link variables $U(x, \mu) = \exp(ag\mathcal{A}_\mu(x))$. As a constraint on S_W the leading term of the expansion into small lattice spacing a should be the continuum action

$$\int d^4x \frac{1}{4} \sum_{\mu, \nu} \sum_d F_{\mu\nu}^d(x) F_{\mu\nu}^d(x), \quad (3.9)$$

where

$$F_{\mu\nu}(x) = \partial_\mu \mathcal{A}_\nu(x) - \partial_\nu \mathcal{A}_\mu(x) + g[\mathcal{A}_\mu(x), \mathcal{A}_\nu(x)], \quad (3.10)$$

with continuum derivatives. For instance, S_W may be the Wilson action

$$S_W(\mathcal{A}; g) = \frac{1}{2g^2} \sum_{\lambda, \mu\nu} \text{tr} [1 - U(x, \mu)U(x + a\hat{\mu}, \nu)U^{-1}(x + a\hat{\nu}, \mu)U^{-1}(x, \nu)]. \quad (3.11)$$

S_{gf} is the gauge fixing contribution

$$S_{\text{gf}}(\mathcal{A}; \lambda) = a^4 \sum_{\lambda, d} \frac{\lambda}{2} (F_d(\mathcal{A}(x)))^2$$

$$= a^4 \sum_{\lambda, d} \frac{\lambda}{2} \left(\frac{1}{a} \sum_\mu \partial_\mu^* \mathcal{A}_\mu^d(x) \right)^2, \quad \lambda > 0 \quad (3.12)$$

and

$$S_{\text{FP}}(\mathcal{A}, c, \bar{c}; g) = a^4 \sum_{\lambda, d} \bar{c}_d(x) (\Delta_{\text{FP}}(\mathcal{A}; g) c)_d(x) \quad (3.13a)$$

is the corresponding Faddeev–Popov action,

$$\begin{aligned}
 & [\Delta_{\text{FP}}(\mathcal{A}; g) c](x) \\
 &= F(s\mathcal{A}(x)) \\
 &= -\frac{1}{a^2} \sum_{\mu} \partial_{\mu}^* \left(\left[\text{ad}(ag\mathcal{A}_{\mu}(x)) + \frac{\text{ad}(ag\mathcal{A}_{\mu}(x))}{1 - \exp[-\text{ad}(ag\mathcal{A}_{\mu}(x))]} \partial_{\mu} \right] c(x) \right). \quad (3.13b)
 \end{aligned}$$

By construction, the measure

$$\mathcal{D}\mathcal{A} \mathcal{D}c \mathcal{D}\bar{c} \exp \left\{ \sum_{x,\mu} \text{tr} \log \frac{1 - \exp[-\text{ad}(ag\mathcal{A}_{\mu}(x))]}{\text{ad}(ag\mathcal{A}_{\mu}(x))} \right\} \quad (3.14)$$

and the action S_0 are invariant under the BRS-transformation (2.21). Note also that S_{gf} is invariant under global gauge transformations

$$\mathcal{A}_{\mu}(x) \rightarrow \mathcal{A}_{\mu}(x) + [\omega, \mathcal{A}_{\mu}(x)], \quad (3.15)$$

where $\omega \in \mathfrak{su}(N)$. All the terms in S_0 are to be understood as formal power series in the gauge field. To a given order in the loop expansion and the lattice spacing only a finite number of vertices must be taken into account. Their number grows with increasing number of loops. Their contributions to the continuum limit however are under control by the WI's which follow from BRS-invariance and which will now be derived. The method is standard. As a consequence of the symmetry, the partition function Z_0 satisfies

$$\sum_{x,d} \left\{ \sum_{\mu} J_{\mu}^d(x) \frac{\partial}{\partial K_{\mu}^d(x)} - \bar{\xi}_d(x) \frac{\partial}{\partial L_d(x)} - a^4 \lambda \xi_d(x) F_d \left(\frac{1}{a^4} \frac{\partial}{\partial J(x)} \right) \right\} Z_0 = 0, \quad (3.16)$$

and the same equation holds for the generating functional $W_0 = \hbar \log Z_0$ of the connected Green functions. A Legendre transformation

$$\begin{aligned}
 & W_0(J, \bar{\xi}, \xi, K, L; g, \lambda) \\
 &= \Gamma_0(\mathcal{A}, c, \bar{c}, K, L; g, \lambda) \\
 &+ a^4 \sum_{x,d} \left\{ \sum_{\mu} J_{\mu}^d(x) \mathcal{A}_{\mu}^d(x) + \bar{\xi}_d(x) c_d(x) + \bar{c}_d(x) \xi_d(x) \right\}, \quad (3.17a)
 \end{aligned}$$

where

$$a^4 \mathcal{A}_\mu^d(x) = \frac{\partial W_0}{\partial J_\mu^d(x)}, \quad a^4 c_d(x) = \frac{\partial W_0}{\partial \xi_d(x)}, \quad a^4 \bar{c}_d(x) = -\frac{\partial W_0}{\partial \xi_d(x)} \quad (3.17b)$$

with inverse relations

$$a^4 J_\mu^d(x) = -\frac{\partial \Gamma_0}{\partial \mathcal{A}_\mu^d(x)}, \quad a^4 \bar{\xi}_d(x) = \frac{\partial \Gamma_0}{\partial c_d(x)}, \quad a^4 \xi_d(x) = -\frac{\partial \Gamma_0}{\partial \bar{c}_d(x)} \quad (3.17c)$$

yields for the vertex functional Γ_0

$$\sum_{\lambda, d} \left(\sum_\mu \frac{\partial \Gamma_0}{\partial \mathcal{A}_\mu^d(x)} \frac{\partial \Gamma_0}{\partial K_\mu^d(x)} + \frac{\partial \Gamma_0}{\partial c_d(x)} \frac{\partial \Gamma_0}{\partial L_d(x)} - a^4 \lambda F_d(\mathcal{A}(x)) \frac{\partial \Gamma_0}{\partial \bar{c}_d(x)} \right) = 0. \quad (3.18)$$

Setting

$$\bar{\Gamma}_0 = \Gamma_0 + a^4 \sum_{\lambda, d} (\lambda/2) (F_d(\mathcal{A}(x)))^2 \quad (3.19)$$

and using the Faddeev–Popov field equations which in terms of $\bar{\Gamma}_0$ reads

$$F_d \left(\frac{\partial \bar{\Gamma}_0}{\partial K(x)} \right) + \frac{\partial \bar{\Gamma}_0}{\partial \bar{c}_d(x)} = 0, \quad (3.20)$$

we get the WI for the vertex functional $\bar{\Gamma}_0$

$$\sum_{\lambda, d} \left(\sum_\mu \frac{\partial \bar{\Gamma}_0}{\partial \mathcal{A}_\mu^d(x)} \frac{\partial \bar{\Gamma}_0}{\partial K_\mu^d(x)} + \frac{\partial \bar{\Gamma}_0}{\partial c_d(x)} \frac{\partial \bar{\Gamma}_0}{\partial L_d(x)} \right) = 0 \quad (3.21)$$

We will show that these identities are stable under renormalization, i.e. they also hold for the renormalized functionals. Note that the differential equation (3.21) is similar in form to the WI for non-abelian continuum gauge field theories [12, 17].

4. Renormalization

Green functions derived from Z_0 are divergent if the cutoff is removed. We are going to prove that as a consequence of the BRS-symmetry they can be made finite by renormalizing the coupling constant and the field strengths, to every finite order

of perturbation theory. That gives the small coupling expansion of the continuum limit of $SU(N)$ lattice field theory in the infinite volume, which is anticipated to exist.

As renormalization scheme we may employ subtractions of overall divergences of one-particle irreducible (1PI) diagrams by normalizing at non-exceptional momenta and by additional finite renormalizations chosen in such a way that in the sum of all diagrams to the given order all IR-singularities drop out. That can be achieved by applying the additional subtractions only to 1PI Green functions with two or three external legs, all being massless, in such a way that they vanish at zero momentum [11]. On the other hand, we are concerned to preserve the symmetry, i.e. we are interested to do subtractions in such a way that renormalized WI's are satisfied, which automatically would insure IR-convergence by the afore-mentioned property. Together with the freedom we have in the choice of the subtraction scheme [9] we are led in a natural way to "minimal renormalizations". That means we employ the minimal set of subtractions (taking locality into account) which insure convergence up to the given order and simultaneously guarantee validity of the WI's for the renormalized amplitudes. As usual, that introduces a mass scale μ which defines the renormalized theory. Afterwards, normalization conditions (at non-exceptional momenta) may be implemented by additional finite contributions to the renormalization constants, hence preserving the WI's. Note that all diagrams which carry vertices resulting from "irrelevant" terms of the lattice action contribute to Green functions in the large cutoff limit only as polynomials in the external momenta of the order of at most the divergence degree, which is the order of subtractions [11]. These contributions are thus absorbed by the normalization. The resulting renormalized theory has the property that the continuum limit does not depend on the specific choice of the gauge field action, and the limit is given by using continuum Feynman integrals and the corresponding normalization conditions.

We write the renormalized partition function Z_R as

$$\begin{aligned} Z_R(J, \bar{\xi}, \xi, K, L; g, \lambda) \\ = N_R \int \mathcal{D}\mathcal{A} \mathcal{D}c \mathcal{D}\bar{c} \exp\{ \mathcal{E}_R(\mathcal{A}, c, \bar{c}, K, L; g, \lambda) + (1/\hbar) S_\ell(\mathcal{A}, c, \bar{c}; J, \bar{\xi}, \xi) \}, \end{aligned} \quad (4.1)$$

where $\mathcal{E}_R(\mathcal{A}, c, \bar{c}, K, L; g, \lambda) = \mathcal{E}_0(\mathcal{A}, c, \bar{c}, K, L, g, \lambda) + O(\hbar^0)$, the source term S_ℓ is as before given by eq. (3.4), and N_R is the normalization factor so that

$$Z_R(\text{all sources} = 0, g, \lambda) = 1.$$

By definition, all basic field Green functions and those containing local operator insertions of the form (3.7) derived from the renormalized partition function Z_R

remain finite if the cutoff is removed, to every order in perturbation theory. Clearly $\mathcal{E}_R$ is not uniquely determined by this condition. In particular, an arbitrary set of irrelevant terms of UV-degree not greater than four and IR-degree not less than four may be added. The following theorem states that renormalization of lattice gauge theory can be traced back to renormalization of the coupling constant and the field strengths.

RENORMALIZATION THEOREM

There exist renormalization constants $Z_g, Z_{\mathcal{A}}, Z_c$, which are formal power series in $\hbar$, so that

$$\mathcal{E}_R(\mathcal{A}, c, \bar{c}, K, L; g, \lambda) = \mathcal{E}_0(Z_{\mathcal{A}}^{1/2}\mathcal{A}, Z_c^{1/2}c, Z_c^{1/2}\bar{c}, Z_c^{1/2}K, Z_{\mathcal{A}}^{1/2}L, Z_g g, Z_{\mathcal{A}}^{-1}\lambda). \quad (4.2)$$

Furthermore, $Z_g, Z_{\mathcal{A}}$ and Z_c depend only logarithmically on the cutoff $\Lambda = a^{-1}$.

If $Z_g, Z_{\mathcal{A}}, Z_c$ are fixed by normalization conditions (rather than minimal subtraction), the continuum limit is universal, i.e. it does not depend on the details of the gauge field action, and the theorem always applies. That is because the renormalization constants depend only logarithmically on the cutoff. Thus, the additional renormalizations to insure (4.2) to all orders in the lattice spacing a are in fact irrelevant, i.e. they vanish in the continuum limit. This would not be the case if the renormalization constants would be higher divergent. Notice that the renormalizations of the gauge field $\mathcal{A}$ and the source L and also of the Faddeev–Popov field c and the source K are equal, respectively. Furthermore, the gauge fixing action (3.12) does not change.

For convenience we introduce the bare quantities

$$\begin{aligned} \phi^0 &= Z_{\phi}^{1/2}\phi, & \phi &= \mathcal{A}, c, \bar{c}, K, L, \\ g_0 &= Z_g g, & \lambda_0 &= Z_{\mathcal{A}}^{-1}\lambda. \end{aligned} \quad (4.3)$$

Then the relation (4.2) may be written as

$$\begin{aligned} \mathcal{E}_R(\mathcal{A}, c, \bar{c}, K, L, g, \lambda) &= \mathcal{E}_0(\mathcal{A}^0, c^0, \bar{c}^0, K^0, L^0; g_0, \lambda_0), \\ Z_{\mathcal{A}} &= Z_L, & Z_c &= Z_{\bar{c}} = Z_K \end{aligned} \quad (4.4)$$

We define in the usual way renormalized functionals of connected and vertex amplitudes by

$$\begin{aligned} W_R &= \hbar \log Z_R, \\ \Gamma_R &= W_R - a^4 \sum_{\nu, d} \left\{ \sum_{\mu} J_{\mu}^d(x) \mathcal{A}_{\mu}^d(x) + \bar{\xi}_d(x) c_d(x) + \bar{c}_d(x) \xi_d(x) \right\}, \end{aligned} \quad (4.5)$$

and

$$\bar{\Gamma}_R = \Gamma_R + a^4 \sum_{x,d} (\lambda/2) (F_d(\mathcal{A}(x)))^2. \quad (4.6)$$

Obviously, the renormalization theorem implies that

$$\bar{\Gamma}_R(\mathcal{A}, c, \bar{c}, K, L; g, \lambda) = \bar{\Gamma}_0(\mathcal{A}^0, c^0, \bar{c}^0, K^0, L^0; g_0, \lambda_0), \quad (4.7)$$

and that the renormalized vertex functional $\bar{\Gamma}_R$ satisfies

$$\sum_{\lambda,d} \left\{ \sum_{\mu} \frac{\partial \bar{\Gamma}_R}{\partial \mathcal{A}_{\mu}^d(x)} \frac{\partial \bar{\Gamma}_R}{\partial K_{\mu}^d(x)} + \frac{\partial \bar{\Gamma}_R}{\partial c_d(x)} \frac{\partial \bar{\Gamma}_R}{\partial L_d(x)} \right\} = 0 \quad (4.8)$$

and

$$F_d \left(\frac{\partial \bar{\Gamma}_R}{\partial K(x)} \right) + \frac{\partial \bar{\Gamma}_R}{\partial \bar{c}_d(x)} = 0. \quad (4.9)$$

Let us remark that already the form of the WI (4.8) indicates universality of the continuum limit of the renormalized lattice gauge theory. The functional $\bar{\Gamma}_R$ exists in the large cutoff limit which means that the coefficient functions of the expansion of $\bar{\Gamma}_R$ in the fields $\mathcal{A}, c, \bar{c}$ and the sources K, L exist. In that limit eq. (4.8) reproduces the well-known WI of the corresponding continuum gauge field theory. That determines the continuum limit up to three parameters (besides the gauge fixing parameter λ) to be fixed by normalization conditions [21]

The renormalization theorem implies that the transformation laws are multiplicatively renormalized. The measure

$$\mathcal{D}\mathcal{A} \mathcal{D}c \mathcal{D}\bar{c} \exp \left\{ \sum_{x,\mu} \text{tr} \log \frac{1 - \exp[-\text{ad}(ag_0 \mathcal{A}_{\mu}^0(x))]}{\text{ad}(ag_0 \mathcal{A}_{\mu}^0(x))} \right\} \quad (4.10)$$

and the renormalized action

$$S_R(\mathcal{A}, c, \bar{c}, g, \lambda) = S_W(\mathcal{A}^0; g_0) + S_{gf}(\mathcal{A}^0; \lambda_0) + S_{FP}(\mathcal{A}^0, c^0, \bar{c}^0; g_0) \quad (4.11)$$

are invariant under the renormalized BRS-transformation

$$\begin{aligned} \delta_{R,\epsilon} \mathcal{A}_{\mu}(x) &= Z_{\mathcal{A}}^{-1/2} \delta_{\epsilon} \mathcal{A}_{\mu}^0(x), \\ \delta_{R,\epsilon} c(x) &= Z_c^{-1/2} \delta_{\epsilon} c^0(x), \\ \delta_{R,\epsilon} \bar{c}(x) &= Z_{\bar{c}}^{-1/2} \delta_{\epsilon} \bar{c}^0(x), \end{aligned} \quad (4.12)$$

where on the right-hand side only bare quantities are used, and δ_ε is defined by (2.21). The same holds for the contributions to $\mathcal{E}_R$ which are linear in K, L . All basic field Green functions and those containing operators

$$-Z_c^{1/2} \frac{1}{a} \left\{ \text{ad} \left(ag_0 \mathcal{A}_\mu^0(x) + \frac{\text{ad}(ag_0 \mathcal{A}_\mu^0(x))}{1 - \exp[-\text{ad}(ag_0 \mathcal{A}_\mu^0(x))]} \partial_\mu \right) c^0(x) \right\} \quad (4.13)$$

and

$$Z_{\mathcal{A}}^{1/2} (g_0/2) \sum_{d,e} c_d^0(x) c_e^0(x) [T_d, T_e] \quad (4.14)$$

are finite.

5. Power counting and auxiliary fields

In order to prove the renormalization theorem we use general ideas [12] which are not specific to the lattice cutoff aside from some specialities to be discussed below. They are based on the presence of a symmetric cutoff. The proof is by induction on the number of loops.

Assuming that the renormalization theorem holds to the order $n-1$ in the loop expansion, the functional $\bar{\Gamma}_R$ which has been renormalized to that order satisfies the identities (4.8, 4.9). From them we shall derive equations for the divergencies in the large cutoff limit which occur at the order n . The statement of the power counting theorem is that overall divergencies are polynomials in momentum space of the order of the divergence degrees of the corresponding diagrams. Their coefficients, i.e. the renormalization constants, are restricted by the WI's. On the lattice the polynomials cannot directly be subtracted because they violate locality and produce new divergencies in higher loop orders. Nevertheless, it is always possible to choose the counterterms as periodic functions by additional irrelevant contributions. Furthermore, from dimensional analysis we may infer that all renormalization constants are at most logarithmically divergent, so that we are allowed to use the freedom in the choice of these finite contributions to arrange that the renormalized action to the order n satisfies (4.2) up to contributions of higher orders, with appropriate n th order renormalization constants, i.e.

$$\begin{aligned} \mathcal{E}_{R,n}(\mathcal{A}, c, \bar{c}, K, L; g, \lambda) \\ = \mathcal{E}_0(\hat{Z}_{\mathcal{A}}^{1/2} \mathcal{A}, \hat{Z}_c^{1/2} c, \hat{Z}_{\bar{c}}^{1/2} \bar{c}, \hat{Z}_K^{1/2} K, \hat{Z}_L^{1/2} L; \hat{Z}_g g, \hat{Z}_{\lambda}^{-1} \lambda) + O(\hbar^{n+1}) \end{aligned} \quad (5.1)$$

The remainder is in fact local and of an UV-degree not greater than four, and we want to subtract it from the action, which would prove the renormalization theorem. The subtraction scheme we have employed is minimal which means that it gives the

smallest set of counterterms which insure finiteness in the large cutoff limit and simultaneously that the renormalized action satisfies (4.2). Normalization conditions may be implemented at every order by finite contributions to the renormalization constants. Without these normalizations, universality of the continuum limit does only hold modulo finite renormalizations.

Unfortunately, there is an obstruction concerning the applicability of the power counting theorem. The general constraints on the structure of the bare action are that it should have a continuum limit and that the IR-degrees of the vertices are not less than four [11]. Both conditions are violated by the measure contribution to (3.5) as it stands. Namely, its leading order term produces a gauge field vertex

$$\text{---}\times\text{---} = \hbar c_R (g^2/a^2), \quad (5.2)$$

which diverges in the limit of vanishing lattice spacing a . Also, it is of an IR-degree two (two massless lines, no momentum). Actually, this technical problem is easily solved by introducing non-propagating Grassman fields $b, \bar{b}$. Let

$$G_\mu(x) = \frac{1 - \exp[-\text{ad}(ag\mathcal{A}_\mu(x))]}{\text{ad}(ag\mathcal{A}_\mu(x))}. \quad (5.3)$$

Then the measure contribution can be written as

$$\begin{aligned} & \exp \left\{ \sum_{\chi, \mu} \text{tr} \log \frac{1 - \exp[-\text{ad}(ag\mathcal{A}_\mu(x))]}{\text{ad}(ag\mathcal{A}_\mu(x))} \right\} \\ & \cong \int \mathcal{D}b \mathcal{D}\bar{b} \exp \left\{ -\frac{1}{\hbar} a^4 \sum_{x, d, \mu} \bar{b}_\mu^d(x) [G_\mu(x) b_\mu(x)]_d \right\}, \end{aligned} \quad (5.4)$$

where

$$\mathcal{D}b \mathcal{D}\bar{b} = \prod_{\chi, d, \mu} db_\mu^d(x) \cdot \prod_{\chi, d, \mu} d\bar{b}_\mu^d(x). \quad (5.5)$$

Now the action has the desired form, so that the loop counting and power counting arguments apply. The introduction of the auxiliary fields $b, \bar{b}$ is only a technical device and should be avoided in explicit calculations. They are massive fields of an IR- and UV-dimension two. The new action also exhibits a BRS-symmetry. If the transformation laws (2.21) are supplemented by

$$\begin{aligned} \delta_\epsilon \bar{b}_\mu^d(x) &= 0, \\ \delta_\epsilon b_\mu^d(x) &= \epsilon sb_\mu^d(x) = -[G_\mu^{-1}(x) \cdot \delta_\epsilon G_\mu(x) \cdot b_\mu(x)]_d, \end{aligned} \quad (5.6)$$

it is easy to see that the product measure

$$\mathcal{D}\mathcal{A} \mathcal{D}c \mathcal{D}\bar{c} \mathcal{D}b \mathcal{D}\bar{b}$$

is invariant under the combined transformations (2.21), (5.6) and that

$$\delta_\epsilon(\bar{b}_\mu(x)G_\mu(x)b_\mu(x)) = 0$$

Furthermore, we get the nilpotence property

$$\delta_\eta \delta_\epsilon b_\mu(x) = 0, \quad (5.7)$$

i.e.

$$s^2 b_\mu(x) = 0.$$

For,

$$\begin{aligned} \delta_\eta \delta_\epsilon b_\mu(x) &= \delta_\eta(-G_\mu^{-1}(x) \cdot \delta_\epsilon G_\mu(x) \cdot b_\mu(x)) \\ &= -(\delta_\eta G_\mu^{-1}(x) \cdot \delta_\epsilon G_\mu(x) + \delta_\epsilon G_\mu^{-1}(x) \cdot \delta_\eta G_\mu(x)) b_\mu(x) \\ &= 0, \end{aligned}$$

where we have used that $\delta_\eta \delta_\epsilon G_\mu(x) = 0$ and $0 = \delta_\eta \delta_\epsilon (G_\mu^{-1}(x)G_\mu(x)) = \delta_\epsilon G_\mu^{-1}(x) \cdot \delta_\eta G_\mu(x) + \delta_\eta G_\mu^{-1}(x) \cdot \delta_\epsilon G_\mu(x)$. Thus, in order to prove the renormalization theorem, we consider from now on the partition function

$$\begin{aligned} \mathcal{Z}_0(J, \bar{\xi}, \xi, \bar{\eta}, \eta, K, L, M; g, \lambda) &= \mathcal{N}_0 \cdot \int \mathcal{D}\mathcal{A} \mathcal{D}c \mathcal{D}\bar{c} \mathcal{D}b \mathcal{D}\bar{b} \\ &\cdot \exp\left\{ \mathcal{B}_0(\mathcal{A}, c, \bar{c}, b, \bar{b}, K, L, M, g, \lambda) + (1/\hbar) \mathcal{S}_c(\mathcal{A}, c, \bar{c}, b, \bar{b}; J, \bar{\xi}, \xi, \bar{\eta}, \eta) \right\}, \end{aligned} \quad (5.8)$$

where $\mathcal{N}_0$ is the normalization

$$\mathcal{Z}_0(\text{all sources} = 0; g, \lambda) = 1 \quad (5.9)$$

and

$$\begin{aligned} \mathcal{S}_c(\mathcal{A}, c, \bar{c}, b, \bar{b}; J, \bar{\xi}, \xi, \bar{\eta}, \eta) &= a^4 \sum_{x,d} \left\{ \sum_\mu \left[J_\mu^d(x) \mathcal{A}_\mu^d(x) + \bar{\eta}_\mu^d(x) b_\mu^d(x) + \bar{b}_\mu^d(x) \eta_\mu^d(x) \right] \right. \\ &\quad \left. + \bar{\xi}_d(x) c_d(x) + \bar{c}_d(x) \xi_d(x) \right\}. \end{aligned} \quad (5.10)$$

Furthermore

$$\begin{aligned}
 \mathcal{B}_0(\mathcal{A}, c, \bar{c}, b, \bar{b}, K, L, M; g, \lambda) \\
 = -(1/\hbar) a^4 \sum_{x, d, \mu} \bar{b}_\mu^d(x) (G_\mu(x) b_\mu(x))_d \\
 + (1/\hbar) a^4 \sum_{x, d} \left\{ \sum_\mu \left[K_\mu^d(x) s \mathcal{A}_\mu^d(x) + M_\mu^d(x) s b_\mu^d(x) \right] + L_d(x) s c_d(x) \right\} \\
 - (1/\hbar) S_0(\mathcal{A}, c, \bar{c}; g, \lambda), \tag{5.11}
 \end{aligned}$$

and S_0 is given by (3.8)ff. The product measure as well as $\mathcal{B}_0$ are invariant under (2.21), (5.6). Let us define $\mathcal{W}_0 = \hbar \log \mathcal{Z}_0$ and

$$\begin{aligned}
 \mathcal{G}_0(\mathcal{A}, c, \bar{c}, b, \bar{b}, K, L, M; g, \lambda) \\
 = \mathcal{W}_0(J, \bar{\xi}, \xi, \bar{\eta}, \eta, K, L, M; g, \lambda) \\
 - a^4 \sum_{x, d} \left\{ \sum_\mu \left[J_\mu^d(x) \mathcal{A}_\mu^d(x) + \bar{\eta}_\mu^d(x) b_\mu^d(x) + \bar{b}_\mu^d(x) \eta_\mu^d(x) \right] \right. \\
 \left. + \bar{\xi}_d(x) c_d(x) + \bar{c}_d(x) \xi_d(x) \right\}, \tag{5.12}
 \end{aligned}$$

$$a^4 \mathcal{A}_\mu^d(x) = \frac{\partial \mathcal{W}_0}{\partial J_\mu^d(x)}, \quad a^4 b_\mu^d(x) = \frac{\partial \mathcal{W}_0}{\partial \bar{\eta}_\mu^d(x)}, \quad a^4 \bar{b}_\mu^d(x) = - \frac{\partial \mathcal{W}_0}{\partial \eta_\mu^d(x)}$$

etc. (cf. eq. (3.17)), and

$$\bar{\mathcal{G}}_0 = \mathcal{G}_0 + a^4 \sum_{x, d} (\lambda/2) (F_d(\mathcal{A}(x)))^2. \tag{5.13}$$

Then $\bar{\mathcal{G}}_0$ satisfies

$$F_d \left(\frac{\partial \bar{\mathcal{G}}_0}{\partial K(x)} \right) + \frac{\partial \bar{\mathcal{G}}_0}{\partial \bar{c}_d(x)} = 0 \tag{5.14}$$

and

$$\sum_{x, d} \left\{ \sum_\mu \left[\frac{\partial \bar{\mathcal{G}}_0}{\partial \mathcal{A}_\mu^d(x)} \frac{\partial \bar{\mathcal{G}}_0}{\partial K_\mu^d(x)} + \frac{\partial \bar{\mathcal{G}}_0}{\partial b_\mu^d(x)} \frac{\partial \bar{\mathcal{G}}_0}{\partial M_\mu^d(x)} \right] + \frac{\partial \bar{\mathcal{G}}_0}{\partial c_d(x)} \frac{\partial \bar{\mathcal{G}}_0}{\partial L_d(x)} \right\}. \tag{5.15}$$

As before one could define $\mathcal{B}_R$ and $\mathcal{Z}_R$ in such a way that all Green functions of

the fields including $b, \bar{b}$ and those containing insertions of $s\mathcal{A}, sc, sb$ are finite if the cutoff is removed. However, the $b, \bar{b}$ -fields have been introduced only for technical reasons, in order to make the measure term consistent with power counting. At the end of the renormalization procedure the source M coupled to the variation of the b -field is set equal to zero, and the integration over the auxiliary fields is done explicitly, giving the renormalized measure (4.10). We are thus not interested in renormalizing Green functions with sb -composite operator insertions. Furthermore, below the WI's will be needed only to determine the structure of the divergencies in $\bar{\mathcal{G}}$ which do not depend on the auxiliary fields. In addition we include a renormalization of the b -field

In all that follows renormalized Green functions mean that they are finite if $M = 0$. We state the

AUXILIARY THEOREM

There are renormalization constants

$$Z_{\mathcal{A}}, Z_c, Z_g, Z_\lambda, Z_K, Z_L, Z_M, Z_b, \quad (5.16)$$

which are at most logarithmically divergent and are formal power series in $\hbar$, so that

$$\begin{aligned} \mathcal{B}_R(\mathcal{A}, c, \bar{c}, b, \bar{b}, K, L, M, g, \lambda) \\ = \mathcal{B}_0(Z_{\mathcal{A}}^{1/2}\mathcal{A}, Z_c^{1/2}c, Z_c^{1/2}\bar{c}, Z_b^{1/2}b, Z_b^{1/2}\bar{b}, Z_K^{1/2}K, Z_L^{1/2}L, Z_M^{1/2}M; Z_g g, Z_\lambda \lambda), \end{aligned} \quad (5.17)$$

and they satisfy the constraints

$$\begin{aligned} Z_{\mathcal{A}} &= Z_L, & Z_c &= Z_K, \\ Z_{\mathcal{A}} Z_c &= Z_M Z_b, & Z_\lambda &= Z_{\mathcal{A}}^{-1}. \end{aligned} \quad (5.18)$$

From this statement the renormalization theorem follows by setting $M = 0$ and doing the $b, \bar{b}$ -integrations, which produce the renormalized measure. The auxiliary field renormalization cancels by the normalization.

As a consequence of (5.17), (5.18), the renormalized vertex functional $\bar{\mathcal{G}}_R$, defined by the analogous steps as $\bar{\mathcal{G}}_0$, satisfies the equations

$$F_d \left(\frac{\partial \bar{\mathcal{G}}_R}{\partial K(x)} \right) + \frac{\partial \bar{\mathcal{G}}_R}{\partial c_d(x)} = 0 \quad (5.19)$$

and

$$\sum_{x,d} \left\{ \sum_{\mu} \left[\frac{\partial \bar{\mathcal{G}}_R}{\partial \mathcal{A}_{\mu}^d(x)} \frac{\partial \bar{\mathcal{G}}_R}{\partial K_{\mu}^d(x)} + \frac{\partial \bar{\mathcal{G}}_R}{\partial b_{\mu}^d(x)} \frac{\partial \bar{\mathcal{G}}_R}{\partial M_{\mu}^d(x)} \right] + \frac{\partial \bar{\mathcal{G}}_R}{\partial c_d(x)} \frac{\partial \bar{\mathcal{G}}_R}{\partial L_d(x)} \right\}. \quad (5.20)$$

6. Proof of the auxiliary theorem

Suppose that the auxiliary theorem has been proved to the order $n-1$ in $\hbar$. We now proceed to show that it also holds to the order n . All 1PI Green functions are at most superficially divergent to that order. The overall divergence degree of a particular diagram γ is given by [9, 11]

$$\omega(\gamma) = 4 + \sum_V (\omega(V) - 4) - N_{\mathcal{A}} - N_c - N_{\bar{c}} - 2N_b - 2N_{\bar{b}}, \quad (6.1)$$

where the sum is over all vertices of γ which correspond to composite operator insertions of $s\mathcal{A}$ or sc , both having $\omega(V) = 2$. (Remember that we do not discuss sb -insertions which would have $\omega(V) = 3$.) We have used the fact that all vertices which correspond to interaction parts of the action have UV-degree four. $N_{\mathcal{A}}$ is the number of external $\mathcal{A}$ -field legs etc. Note that the IR- and UV-degrees for every field coincide. In particular, we inferred from (5.11) that the auxiliary $b, \bar{b}$ -fields have UV- and IR-degree equal to two. Hence all interaction vertices have an IR-degree not less than four. Together with the WI's that insures IR-convergence. $\omega(\gamma)$ is the same for all γ 's contributing to the same Green function independent of whether they contain irrelevant vertices or not. By power counting, in momentum space the overall divergencies are polynomials of the order of the divergence degree. Expanding $\bar{\mathcal{G}}_{R, n-1}$, the vertex functional which is assumed to be renormalized to the order $n-1$, up to the order n , we get for large cutoff from (5.20)

$$\begin{aligned} S(\bar{\mathcal{G}}_{\text{div}}) = \sum_{x,d} \left\{ \sum_{\mu} \left[\frac{\partial \bar{\mathcal{B}}_0}{\partial \mathcal{A}_{\mu}^d(x)} \frac{\partial \bar{\mathcal{G}}_{\text{div}}}{\partial K_{\mu}^d(x)} + \frac{\partial \bar{\mathcal{B}}_0}{\partial K_{\mu}^d(x)} \frac{\partial \bar{\mathcal{G}}_{\text{div}}}{\partial \mathcal{A}_{\mu}^d(x)} \right. \right. \\ \left. \left. + \frac{\partial \bar{\mathcal{B}}_0}{\partial b_{\mu}^d(x)} \frac{\partial \bar{\mathcal{G}}_{\text{div}}}{\partial M_{\mu}^d(x)} + \frac{\partial \bar{\mathcal{B}}_0}{\partial M_{\mu}^d(x)} \frac{\partial \bar{\mathcal{G}}_{\text{div}}}{\partial b_{\mu}^d(x)} \right] \right. \\ \left. + \frac{\partial \bar{\mathcal{B}}_0}{\partial c_d(x)} \frac{\partial \bar{\mathcal{G}}_{\text{div}}}{\partial L_d(x)} + \frac{\partial \bar{\mathcal{B}}_0}{\partial L_d(x)} \frac{\partial \bar{\mathcal{G}}_{\text{div}}}{\partial c_d(x)} \right\} = O(M), \quad (6.2) \end{aligned}$$

where $\bar{\mathcal{G}}_{\text{div}}$ denotes the n th order divergence. $\hbar \bar{\mathcal{B}}_0$ is the lowest order contribution

to $\bar{\mathcal{G}}_{R, n-1}$, i.e.

$$\hbar \bar{\mathcal{B}}_0 = \hbar \mathcal{B}_0 + (\lambda/2) a^4 \sum_{\mathbf{x}, d} (F_d(\mathcal{A}(x)))^2, \quad (6.3)$$

and $\mathcal{B}_0$ is given by eq. (5.11). Furthermore, due to the linearity of the gauge fixing function, $\bar{\mathcal{G}}_{\text{div}}$ satisfies the equations

$$F_d \left(\frac{\partial \bar{\mathcal{G}}_{\text{div}}}{\partial K(x)} \right) + \frac{\partial \bar{\mathcal{G}}_{\text{div}}}{\partial \bar{c}_d(x)} = 0. \quad (6.4)$$

The most general form of $\bar{\mathcal{G}}_{\text{div}}$ is determined by power counting, and the identities (6.2, 6.4) further restrict the possible coefficients. We cannot simply subtract $\bar{\mathcal{G}}_{\text{div}}$ from $\bar{\mathcal{G}}_{R, n-1}$ because the resulting action would be non-local. Nevertheless, if in the momentum space representation of $\bar{\mathcal{G}}_{\text{div}}$ the momenta are replaced by “lattice momenta” [9], i.e. periodic functions which for small lattice spacing a reproduce the momenta, we get an allowed counterterm which is periodic with the Brillouin zone. In configuration space that means we have to employ lattice derivatives. We state the following

LEMMA

The counterterm $\bar{\mathcal{G}}_{\text{CT}}$ which subtracts the n th order divergencies and satisfies $S(\bar{\mathcal{G}}_{\text{CT}}) = \text{finite} + O(M)$ is given by

$$\begin{aligned} \bar{\mathcal{G}}_{\text{CT}} = & -a^4 \sum_{\mathbf{x}} \text{tr} \left\{ \sum_{\mu} (K_{\mu}(x) - \bar{c}(x) \partial_{\mu}^*) (-1/a) (z_1 \text{ad}(ag \mathcal{A}_{\mu}(x)) + z_2 \partial_{\mu}) c(x) \right. \\ & \left. + z_3 L(x) g(c(x))^2 + r_b \sum_{\mu} \bar{b}_{\mu}(x) b_{\mu}(x) + H(\mathcal{A}(x)) \right\}, \quad (6.5) \end{aligned}$$

where

$$\begin{aligned} H(\mathcal{A}(x)) = & -\frac{1}{4} \sum_{\mu, \nu} \left\{ r_{\mathcal{A}} (1/a^2) (\partial_{\mu} \mathcal{A}_{\nu}(x) - \partial_{\nu} \mathcal{A}_{\mu}(x))^2 \right. \\ & + r_3 2(g/a) (\partial_{\mu} \mathcal{A}_{\nu}(x) - \partial_{\nu} \mathcal{A}_{\mu}(x)) \\ & \left. \times [\mathcal{A}_{\mu}(x), \mathcal{A}_{\nu}(x)] + r_4 g^2 [\mathcal{A}_{\mu}(x), \mathcal{A}_{\nu}(x)]^2 \right\}. \quad (6.6) \end{aligned}$$

Furthermore, the coefficients $z_1, z_2, z_3, r_{\mathcal{A}}, r_3, r_4, r_b$ are at most logarithmically

divergent and satisfy the relations

$$z_1 = z_3, \quad z_2 - z_1 = r_{\mathcal{A}} - r_3 = r_3 - r_4. \quad (6.7)$$

All fields have been written as Lie algebra valued objects, e.g.

$$K_\mu(x) = \sum_d K_\mu^d(x) T_d,$$

and we assume the normalization (2.2). The trace is with respect to the Lie algebra $\mathfrak{su}(N)$. ∂_μ and ∂_μ^* are the lattice derivatives (2.6). $\bar{\mathcal{G}}_{\text{CT}}$ subtracts the n th order divergencies from $\bar{\mathcal{G}}_{\text{R}, n-1}$ and is local. The lemma will be proved at the end of the section.

The constraints (6.7) on the coefficients allow us to write $\bar{\mathcal{G}}_{\text{CT}}$ as the variation of a “continuum-like” action. If we introduce the renormalizations of the fields, the sources and the coupling constant, i.e. if we set

$$\begin{aligned} z_1 &= r_g + \frac{1}{2}r_{\mathcal{A}} + \frac{1}{2}r_K + \frac{1}{2}r_c, \\ z_2 &= \frac{1}{2}r_K + \frac{1}{2}r_c, \end{aligned} \quad (6.8)$$

we get

$$r_3 = \frac{3}{2}r_{\mathcal{A}} + r_g, \quad r_4 = \frac{4}{2}r_{\mathcal{A}} + 2r_g, \quad (6.9)$$

which determines $\bar{\mathcal{G}}_{\text{CT}}$ to be of the form

$$\begin{aligned} \bar{\mathcal{G}}_{\text{CT}} &= \left\{ \frac{r_{\mathcal{A}}}{2} \frac{\partial}{\partial \tau_{\mathcal{A}}} + \frac{r_c}{2} \left(\frac{\partial}{\partial \tau_c} + \frac{\partial}{\partial \tau_{\bar{c}}} \right) + r_g \frac{\partial}{\partial \tau_g} + \frac{r_K}{2} \frac{\partial}{\partial \tau_K} + \frac{r_L}{2} \frac{\partial}{\partial \tau_L} + \frac{r_b}{2} \left(\frac{\partial}{\partial \tau_b} + \frac{\partial}{\partial \tau_{\bar{b}}} \right) \right\} \\ &\quad \bar{\mathcal{G}}_c(\tau_{\mathcal{A}}\mathcal{A}, \tau_c c, \tau_{\bar{c}}\bar{c}, \tau_b b, \tau_{\bar{b}}\bar{b}, \tau_K K, \tau_L L, \tau_g g) \Big|_{\text{all } \tau=1}, \end{aligned} \quad (6.10)$$

where

$$\begin{aligned} &\bar{\mathcal{G}}_c(\mathcal{A}, c, \bar{c}, b, \bar{b}, K, L; g) \\ &= -a^4 \sum_x \text{tr} \left\{ \sum_\mu (K_\mu(x) - \bar{c}(x) \partial_\mu^*) (-1/a) (\text{ad}(a g_{\mathcal{A}} \mathcal{A}_\mu(x)) + \partial_\mu) c(x) \right. \\ &\quad \left. + L(x) g(c(x))^2 + \sum_\mu \bar{b}_\mu(x) b_\mu(x) - \frac{1}{4} \sum_{\mu, \nu} (F_{\mu\nu}(x))^2 \right\}. \end{aligned} \quad (6.11)$$

$F_{\mu\nu}$ is given by eq. (3.10), and the coefficients satisfy the relations

$$r_K = r_c, \quad r_L = r_{\mathcal{A}}. \quad (6.12)$$

$\bar{\mathcal{G}}_c$ differs from $\hbar \bar{\mathcal{B}}_0$ only by irrelevant vertices. Using the fact that all constants r_K , r_L , r_g and r_b are at most logarithmically cutoff dependent, we add to $\bar{\mathcal{G}}_{CT}$ irrelevant contributions in such a way that we get a new counterterm of the form

$$\begin{aligned} \bar{\mathcal{G}}'_{CT} = & \left\{ \frac{r_{\mathcal{A}}}{2} \frac{\partial}{\partial \tau_{\mathcal{A}}} + \frac{r_c}{2} \left(\frac{\partial}{\partial \tau_c} + \frac{\partial}{\partial \tau_{\bar{c}}} \right) + r_g \frac{\partial}{\partial \tau_g} \right. \\ & \left. + \frac{r_K}{2} \frac{\partial}{\partial \tau_K} + \frac{r_L}{2} \frac{\partial}{\partial \tau_L} + r_b \left(\frac{\partial}{\partial \tau_b} + \frac{\partial}{\partial \tau_{\bar{b}}} \right) + \frac{r_M}{2} \frac{\partial}{\partial \tau_M} \right\} \\ & \hbar \bar{\mathcal{B}}_0 \left(\tau_{\mathcal{A}} \mathcal{A}, \tau_c c, \tau_{\bar{c}} \bar{c}, \tau_b b, \tau_{\bar{b}} \bar{b}, \tau_K K, \tau_L L, \tau_M M; \tau_g g \right) \Big|_{\text{all } \tau=1}. \end{aligned} \quad (6.13)$$

Here for convenience we have introduced a logarithmically divergent constant r_M , i.e. a finite renormalization of sb , which will be chosen appropriately below. The additional terms introduced in eq. (6.13) will modify higher order counterterms. However, the power counting is not invalidated by such irrelevant terms. Green functions are always made finite by subtractions of the order of the divergence degrees which do not change. Finally, using the induction hypothesis that the action which gives finite Green functions up to the order $n-1$ in $\hbar$ satisfies

$$\begin{aligned} & \mathcal{B}_{R,n-1}(\mathcal{A}, c, \bar{c}, b, \bar{b}, K, L, M; g, \lambda) \\ &= \mathcal{B}_0 \left(Z_{\mathcal{A}}^{1/2} \mathcal{A}, Z_c^{1/2} c, Z_{\bar{c}}^{1/2} \bar{c}, Z_b^{1/2} b, Z_{\bar{b}}^{1/2} \bar{b}, Z_K^{1/2} K, Z_L^{1/2} L, Z_M^{1/2} M; Z_g g, Z_\lambda \lambda \right), \\ & Z_{\mathcal{A}} = Z_L, \quad Z_c = Z_K, \quad Z_{\mathcal{A}} Z_c = Z_M Z_b, \quad Z_\lambda = Z_{\mathcal{A}}^{-1}, \end{aligned} \quad (6.14)$$

we get

$$\begin{aligned} & \mathcal{B}_{R,n-1} + (1/\hbar) \bar{\mathcal{G}}'_{CT} \\ &= \mathcal{B}_0 \left(\hat{Z}_{\mathcal{A}}^{1/2} \mathcal{A}, \hat{Z}_c^{1/2} c, \hat{Z}_{\bar{c}}^{1/2} \bar{c}, \hat{Z}_b^{1/2} b, \hat{Z}_{\bar{b}}^{1/2} \bar{b}, \hat{Z}_K^{1/2} K, \hat{Z}_L^{1/2} L, \hat{Z}_M^{1/2} M; \hat{Z}_g g, \hat{Z}_\lambda \lambda \right) \\ &+ (1/\hbar) \Delta, \end{aligned} \quad (6.15)$$

where

$$\begin{aligned}\hat{Z}_\phi &= Z_\phi + r_\phi, \quad \phi = \mathcal{A}, c, b, K, L, M, g, \\ \hat{Z}_\lambda &= \hat{Z}_{\mathcal{A}}^{-1}.\end{aligned}\tag{6.16}$$

Note that the gauge fixing part of the action is not changed. Choosing

$$r_M = \frac{Z_{\mathcal{A}} r_c + r_{\mathcal{A}} Z_c + r_{\mathcal{A}} r_c - Z_M r_b}{Z_b + r_b} = r_c + r_{\mathcal{A}} - r_b + O(\hbar^{n+1}),\tag{6.17}$$

the renormalization constants satisfy

$$\hat{Z}_{\mathcal{A}} = \hat{Z}_L, \quad \hat{Z}_c = \hat{Z}_K, \quad \hat{Z}_{\mathcal{A}} \hat{Z}_c = \hat{Z}_M \hat{Z}_b.\tag{6.18}$$

Obviously, the coefficients of the remainder Δ are at most logarithmically divergent in the large cutoff limit. Furthermore, Δ is of the order $\hbar^{n+1}$, local and has UV-degree four and IR-degree not less than four. Thus we are allowed to subtract it from the action $\mathcal{B}_{R, n-1} + \hbar^{-1} \bar{\mathcal{G}}'_{CT}$ (without invalidating power counting in higher orders). That yields the action

$$\mathcal{B}_{R, n} = \mathcal{B}_{R, n-1} + (1/\hbar) \bar{\mathcal{G}}'_{CT} - (1/\hbar) \Delta,\tag{6.19}$$

which is renormalized to the order in question and satisfies (5.17), where the Z are to be replaced by $\hat{Z}$. Thus the induction is closed, and the auxiliary theorem is proved.

We still have to prove the lemma. Only diagrams with non-negative overall divergence degrees are to be discussed. Let us first consider Green functions with insertions of $s\mathcal{A}$ and sc . These composite operators have UV-degree two. Due to ghost number conservation $\bar{\mathcal{G}}'_{CT}$ is at most linear K and L . We shall use invariance of the theory under the global, linear transformation

$$\mathcal{P}(x) \rightarrow [\omega, \mathcal{P}(x)],\tag{6.20}$$

where $\mathcal{P}$ collects all fields $\mathcal{A}, c, \bar{c}, b, \bar{b}$ and sources K, L, M of the composite operators involved. $\omega \in \mathfrak{su}(N)$ is a space-time independent infinitesimal quantity. (6.20) is the infinitesimal version of

$$\mathcal{P}(x) \rightarrow \Lambda \mathcal{P}(x) \Lambda^{-1},\tag{6.21}$$

where $\Lambda \in \mathrm{SU}(N)$. Such a linear symmetry is preserved under renormalization, and

the representations under which the fields transform do not change [12]. Hence the counterterm proportional to K , L is of the form

$$\begin{aligned} \bar{\mathcal{G}}_{\text{CT}, KL} = a^4 \sum_{\lambda, d} \left\{ \sum_{\mu} K_{\mu}^d(x) (-1/a) \left[\left(z_1 \text{ad} \left(ag \mathcal{A}_{\mu}(x) \right) + z_2 \partial_{\mu} \right) c(x) \right]_d \right. \\ \left. + L_d(x) z_3 g \left(c^2(x) \right)_d \right\}. \quad (6.22) \end{aligned}$$

From dimensional analysis we see that z_1, z_2, z_3 are only logarithmically cutoff dependent. Inserting (6.22) into the WI (6.2), the term linear in K and L yields

$$z_1 = z_3 \quad (6.23)$$

Next we use the Faddeev–Popov equations of motion (6.4). We infer that the contribution to $\bar{\mathcal{G}}_{\text{CT}}$ which depends on $\bar{c}$ is given by

$$\bar{\mathcal{G}}_{\text{CT}, \bar{c}} = a^4 \sum_{\lambda, d} \bar{c}_d(x) (1/a^2) \partial_{\mu}^* \left[\left(z_1 \text{ad} \left(ag \mathcal{A}_{\mu}(x) \right) + z_2 \partial_{\mu} \right) c(x) \right]_d. \quad (6.24)$$

In particular, no mass term for the Faddeev–Popov fields is needed, and Green functions with more than two external ghost fields legs are superficially convergent, although for instance the four-point function

$$\left\langle \bar{c}_d(x) \bar{c}_e(y) c_f(z) c_h(w) \right\rangle_{\text{1PI}}$$

has an overall divergence degree equal to zero.

Further candidates of superficially divergent diagrams are those with two external auxiliary field legs

$$\left\langle \bar{b}_{\mu}^d(x) b_{\nu}^e(y) \right\rangle_{\text{1PI}},$$

having divergence degree zero. Thus, we include a contribution

$$a^4 \sum_{\lambda, d} \sum_{\mu} r_b \bar{b}_{\mu}^d(x) b_{\mu}^d(x) \quad (6.25)$$

into $\bar{\mathcal{G}}_{\text{CT}}$. r_b is at most logarithmically divergent. What is left to be discussed are Green functions which have only gauge field legs. The most general form of these

terms is given by

$$\begin{aligned}
 & -a^4 \sum_x H(\mathcal{A}(x)) \\
 & = a^4 \sum_x \frac{1}{4} \sum_{\mu, \nu} \text{tr} \left[r_{\mathcal{A}} \frac{1}{a^2} \left(\partial_{\mu} \mathcal{A}_{\nu}(x) - \partial_{\nu} \mathcal{A}_{\mu}(x) \right)^2 + r_{\lambda} \frac{1}{a^2} \partial_{\mu} \mathcal{A}_{\mu}(x) \partial_{\nu} \mathcal{A}_{\nu}(x) \right. \\
 & \quad + 2gr_3 \frac{1}{a} \left(\partial_{\mu} \mathcal{A}_{\nu}(x) - \partial_{\nu} \mathcal{A}_{\mu}(x) \right) \left[\mathcal{A}_{\mu}(x), \mathcal{A}_{\nu}(x) \right] \\
 & \quad + 2g\beta \frac{1}{a} \partial_{\mu} \mathcal{A}_{\nu}(x) \{ \mathcal{A}_{\mu}(x), \mathcal{A}_{\nu}(x) \} \\
 & \quad \left. + g^2 r_4 \left[\mathcal{A}_{\mu}(x), \mathcal{A}_{\nu}(x) \right]^2 + g^2 \gamma \{ \mathcal{A}_{\mu}(x), \mathcal{A}_{\nu}(x) \}^2 \right], \quad (6.26)
 \end{aligned}$$

where $[,]$ denotes the commutator and $\{ , \}$ the anti-commutator. We have used that the two- and three-point functions vanish at zero momentum, as a consequence of the WI's (cf. appendix A or refs. [22, 23]). All other counterterms which respect the global symmetries and thus could contribute to eq. (6.26) actually have vanishing coefficients as can easily be seen by inserting them into the WI (6.2). That is because there is no term from the variation of the tree level gauge field action (the first term in eq. (6.2)) which could compensate such terms. Again dimensional analysis shows that the coefficients $r_{\mathcal{A}}, r_{\lambda}, r_3, r_4, \beta, \gamma$ are at most logarithmically cutoff dependent. Inserting eq. (6.26) into the WI (6.2) and using that for small lattice spacing a the lattice gauge field action reduces to

$$-a^4 \sum_x \frac{1}{4} \sum_{\mu, \nu} \text{tr} F_{\mu\nu}^2(x), \quad (6.27)$$

where $F_{\mu\nu}$ is given by eq. (3.10), we get after some tedious algebra

$$\begin{aligned}
 r_{\lambda} &= \beta = \gamma = 0, \\
 r_3 - r_4 &= r_{\mathcal{A}} - r_3 = z_2 - z_1. \quad (6.28)
 \end{aligned}$$

That exhausts all possible superficially divergent diagrams to the order n we are interested in. Thus, $\bar{\mathcal{G}}_{\text{CT}}$ is the sum of the terms (6.22), (6.24), (6.25) and (6.26), where the coefficients are constrained by (6.23) and (6.28). That proves the lemma.

7. Gauge independence of the β -function

It is well known that for dimensionally regularized gauge theories the β -function derived from the coupling constant renormalization is independent of the gauge fixing, at least in the minimal renormalization [23]. We will show here that this conventional result also holds for the lattice cutoff. More precisely, we prove that for the Lorentz gauge

$$F_d(\mathcal{A}(x)) = \sum_{\mu} (1/a) \partial_{\mu}^* \mathcal{A}_{\mu}^d(x), \quad (7.1)$$

the coupling constant renormalization Z_g , defined by

$$g_0 = Z_g(g, \lambda, \mu a) g, \quad (7.2)$$

where g_0 is the bare coupling constant (cf. eq. (4.3)), does not depend on the parameter λ of the gauge fixing action (3.12), if the minimal subtraction scheme is chosen. That means all renormalization constants of the renormalized action (4.2) are asymptotic power series in $\log \mu a$ with leading order term equal to 1. Z_g is given by

$$Z_g = 1 + \sum_{n \geq 1} r_g^{(n)}, \quad (7.3)$$

where $r^{(n)}$ is the n th order term in $\hbar$ as defined in the preceding section and depends logarithmically on the cutoff. It may be written as the difference between the n th order divergencies of the vertex functions with three and two gauge field legs (cf. (6.9))

$$r_g^{(n)} = r_3^{(n)} - \frac{3}{2} r_{\mathcal{A}}^{(n)}. \quad (7.4)$$

We intend to determine the dependence of these coefficients on the gauge, and we will see that the right-hand side of (7.4) is λ -independent, which proves the assertion. The proof is based on the observation that an infinitesimal variation of the gauge fixing action can be represented as a BRS-variation of an “integrated” local operator of IR- and UV-degree three which carries the Faddeev–Popov charge of the anti-ghost [17, 23] (remember the transformation law of $\bar{c}$). One such operator may always be inserted into Green functions without destroying their IR-convergence (in four dimensions) [6, 11]. It should be coupled to a space–time independent Grassman source. The variation of Green functions under a change of the gauge written in terms of such an operator insertion gives rise to a new set of WTs. We derive them for the bare amplitudes first. Then using multiplicative renormalizability, which holds also for the new operator (with the exception of eq. (7.8) below), we easily get similar equations for the renormalized functions, from which we infer the λ -independence of Z_g , order by order in perturbation theory.

Let us introduce into the action $\mathcal{E}_0$ as given by eq. (3.5) the composite operator

$$\frac{1}{\hbar} a^4 N \sum_{x, d, \mu} \left[\bar{c}_d(x) \frac{1}{a} \partial_\mu^* \mathcal{A}_\mu^d(x) - K_\mu^d(x) \mathcal{A}_\mu^d(x) \right] \quad (7.5)$$

where N is a space-time independent Grassman source, $N^2 = 0$, anticommuting with $c, \bar{c}, K$. Notice that a BRS-variation of the first term mainly reproduces the gauge fixing action and an operator that can be dealt with by the Faddeev–Popov field equations. The second term has been introduced in order to get finite Green functions with simultaneous insertions of

$$a^4 \sum_{x, d, \mu} \bar{c}_d(x) \frac{1}{a} \partial_\mu^* \mathcal{A}_\mu^d(x) \quad (7.6)$$

and

$$s \mathcal{A}_\mu^d(x) \quad (7.7)$$

(cf. eq. (3.7)), by the same renormalization prescription as in sect 4, supplemented by a renormalization of the source N . Similarly, we should introduce a term

$$\gamma N a^4 \sum_{x, d} L_d(x) c_d(x), \quad (7.8)$$

with a renormalization constant γ , which depends logarithmically on the cutoff. However, we do not need it for our purposes, only operator insertions (7.6) and (7.7) are to be discussed below. Actually, that the operator (7.5) is multiplicatively renormalized (except (7.8)) follows from the Faddeev–Popov field equations, as shown in appendix B.

Let us write unnormalized expectation values as

$$[O]_J, \quad [O]_0 = [O]_{J=0}, \quad (7.9)$$

where $J = (J, \bar{\xi}, \xi, K, L, N)$, so that

$$Z_0(J) = [1]_J / [1]_0. \quad (7.10)$$

Then we have

$$\lambda \frac{\partial}{\partial \lambda} W_0(J) = \lambda \frac{\partial}{\partial \lambda} \hbar \log Z_0(J) = \frac{[-S_{\text{gf}}]_J}{[1]_J} - \frac{[-S_{\text{gf}}]_0}{[1]_0}, \quad (7.11)$$

where S_{gf} is given by eq. (3.12) Eq (7.11) will be written as a differential equation

for W_0 . A BRS-transformation (2.21) applied to $Z_0(J)$ yields

$$\begin{aligned}
 0 = \sum_{x,d} \left\{ \sum_{\mu} \left(J_{\mu}^d(x) - N K_{\mu}^d(x) \right) \left(\frac{\partial}{\partial K_{\mu}^d(x)} - N \frac{\partial}{\partial J_{\mu}^d(x)} \right) \right. \\
 \left. - \bar{\xi}_d(x) \frac{\partial}{\partial L_d(x)} - \xi_d(x) \lambda F_d \left(\frac{\partial}{\partial J(x)} \right) \right\} [1], \\
 + N \left[2S_{\text{gf}} + a^4 \sum_{x,d} \bar{c}_d(x) (\Delta_{\text{FPC}})_d(x) \right], \quad (7.12)
 \end{aligned}$$

Furthermore, differentiating the Faddeev–Popov equations of motion as stated in appendix B, eq (B 1) with respect to the source ξ , we get

$$\begin{aligned}
 0 = \left\{ \frac{1}{[1]}, \frac{\partial}{\partial \xi_d(x)} (\text{B 1}) \right\} - \left\{ \frac{1}{[1]}, \frac{\partial}{\partial \xi_d(x)} (\text{B 1}) \right\}_{j=0} \\
 = \frac{1}{[1]}, \left[a^4 \bar{c}_d(x) (\Delta_{\text{FPC}})_d(x) \right]_j - \frac{1}{[1]_0} \left[a^4 \bar{c}_d(x) (\Delta_{\text{FPC}})_d(x) \right]_0 \\
 - \frac{1}{[1]}, \xi_d(x) \frac{\partial}{\partial \xi_d(x)} [1]_j + O(N) \quad (7.13)
 \end{aligned}$$

Inserting eqs. (7.12) and (7.13) into (7.11) yields

$$\begin{aligned}
 2N\lambda \frac{\partial}{\partial \lambda} W_0(J) = \sum_{x,d} \left\{ \sum_{\mu} \left(J_{\mu}^d(x) - N K_{\mu}^d(x) \right) \left(\frac{\partial}{\partial K_{\mu}^d(x)} - N \frac{\partial}{\partial J_{\mu}^d(x)} \right) \right. \\
 \left. - \bar{\xi}_d(x) \frac{\partial}{\partial L_d(x)} + N \xi_d(x) \frac{\partial}{\partial \xi_d(x)} - \xi_d(x) \lambda F_d \left(\frac{\partial}{\partial J(x)} \right) \right\} W_0(J). \quad (7.14)
 \end{aligned}$$

Applying a Legendre transformation as in eq. (3.17) and expanding the vertex functional Γ_0 in powers of N ,

$$\Gamma_0 = \Gamma + N(\partial/\partial N)\Gamma, \quad (7.15)$$

the N -independent term reproduces the WI (3.18), whereas the term linear in N yields

$$\begin{aligned}
 & 2\lambda \frac{\partial}{\partial \lambda} \Gamma \\
 &= a^4 \sum_{x,d} \left\{ \sum_{\mu} \left(\frac{1}{a^4} \frac{\partial \Gamma}{\partial \mathcal{A}_{\mu}^d(x)} \mathcal{A}_{\mu}^d(x) - K_{\mu}^d(x) \frac{1}{a^4} \frac{\partial \Gamma}{\partial K_{\mu}^d(x)} \right) - \bar{c}_d(x) \frac{1}{a^4} \frac{\partial \Gamma}{\partial \bar{c}_d(x)} \right. \\
 &\quad - \frac{\partial}{\partial N} \left[\sum_{\mu} \frac{1}{a^4} \frac{\partial \Gamma}{\partial \mathcal{A}_{\mu}^d(x)} \frac{1}{a^4} \frac{\partial \Gamma}{\partial K_{\mu}^d(x)} + \frac{1}{a^4} \frac{\partial \Gamma}{\partial c_d(x)} \frac{1}{a^4} \frac{\partial \Gamma}{\partial L_d(x)} \right. \\
 &\quad \left. \left. - \lambda \sum_{\mu} \frac{1}{a} \partial_{\mu}^* \mathcal{A}_{\mu}^d(x) \frac{1}{a^4} \frac{\partial \Gamma}{\partial \bar{c}_d(x)} \right] \right\}. \quad (7.16)
 \end{aligned}$$

Finally, setting

$$\bar{\Gamma} = \Gamma + a^4 \sum_{x,d} (\lambda/2) (F_d(\mathcal{A}(x)))^2, \quad (7.17)$$

we get the desired differential equation which describes the explicit λ -dependence of the vertex functions:

$$\begin{aligned}
 & 2\lambda \frac{\partial}{\partial \lambda} \bar{\Gamma} \\
 &= a^4 \sum_{x,d} \left\{ \sum_{\mu} \left(\frac{1}{a^4} \frac{\partial \bar{\Gamma}}{\partial \mathcal{A}_{\mu}^d(x)} \mathcal{A}_{\mu}^d(x) - K_{\mu}^d(x) \frac{1}{a^4} \frac{\partial \bar{\Gamma}}{\partial K_{\mu}^d(x)} \right) - \bar{c}_d(x) \frac{1}{a^4} \frac{\partial \bar{\Gamma}}{\partial \bar{c}_d(x)} \right. \\
 &\quad \left. - \frac{\partial}{\partial N} \left(\sum_{\mu} \frac{1}{a^4} \frac{\partial \bar{\Gamma}}{\partial \mathcal{A}_{\mu}^d(x)} \frac{1}{a^4} \frac{\partial \bar{\Gamma}}{\partial K_{\mu}^d(x)} + \frac{1}{a^4} \frac{\partial \bar{\Gamma}}{\partial c_d(x)} \frac{1}{a^4} \frac{\partial \bar{\Gamma}}{\partial L_d(x)} \right) \right\}. \quad (7.18)
 \end{aligned}$$

Differentiation with respect to the gauge fields yields the λ -dependence of the two- and three-point functions we are interested in (zero sources are assumed at the end of the differentiation):

$$\begin{aligned}
 & 2\lambda \frac{\partial}{\partial \lambda} \frac{1}{(a^4)^2} \frac{\partial^2 \bar{\Gamma}}{\partial \mathcal{A}_{\rho}^h(z) \partial \mathcal{A}_{\nu}^e(y)} \\
 &= 2 \frac{1}{(a^4)^2} \frac{\partial^2 \bar{\Gamma}}{\partial \mathcal{A}_{\rho}^h(z) \partial \mathcal{A}_{\nu}^e(y)} \\
 &\quad - \frac{1}{(a^4)^3} \sum_{x,d,\mu} \left(\frac{\partial^2 \bar{\Gamma}}{\partial \mathcal{A}_{\nu}^e(y) \partial \mathcal{A}_{\mu}^d(x)} \frac{\partial^3 \bar{\Gamma}}{\partial \mathcal{A}_{\rho}^h(z) \partial N \partial K_{\mu}^d(x)} + 1 \text{ perm} \right) \quad (7.19)
 \end{aligned}$$

and

$$\begin{aligned}
 & 2\lambda \frac{\partial}{\partial \lambda} \frac{1}{(a^4)^3} \frac{\partial^3 \bar{\Gamma}}{\partial \mathcal{A}_\sigma^h(w) \partial \mathcal{A}_\rho^m(z) \partial \mathcal{A}_\nu^e(y)} \\
 &= 3 \frac{1}{(a^4)^3} \frac{\partial^3 \bar{\Gamma}}{\partial \mathcal{A}_\sigma^h(w) \partial \mathcal{A}_\rho^m(z) \partial \mathcal{A}_\nu^e(y)} \\
 &\quad - \frac{1}{(a^4)^4} \sum_{x,d,\mu} \left\{ \frac{\partial^3 \bar{\Gamma}}{\partial \mathcal{A}_\rho^m(z) \partial \mathcal{A}_\nu^e(y) \partial \mathcal{A}_\mu^d(x)} \frac{\partial^3 \bar{\Gamma}}{\partial \mathcal{A}_\sigma^h(w) \partial N \partial K_\mu^d(x)} \right. \\
 &\quad \quad \quad + 2 \text{ perm} \\
 &\quad \quad \quad \left. + \frac{\partial^2 \bar{\Gamma}}{\partial \mathcal{A}_\nu^e(y) \partial \mathcal{A}_\mu^d(x)} \frac{\partial^4 \bar{\Gamma}}{\partial \mathcal{A}_\sigma^h(w) \partial \mathcal{A}_\rho^m(z) \partial N \partial K_\mu^d(x)} + 2 \text{ perm} \right\}. \quad (7.20)
 \end{aligned}$$

In the following we use the short-hand notation

$$\bar{\Gamma}^{(2\mathcal{A}, N, K)}(x, \alpha, b; y, \beta, c | z, \gamma, d) = \frac{\partial^4 \bar{\Gamma}}{\partial \mathcal{A}_\alpha^h(x) \partial \mathcal{A}_\beta^e(y) \partial N \partial K_\gamma^d(z)} \Big|_{\mathcal{A}=0} \quad = N=0 \quad (7.21)$$

and similar expressions for the other functions.

Note that so far all parameters are bare ones. Now assume that renormalization has been carried out up to the order n (indicated by a subscript R), and that the coupling constant renormalization Z_g to this order is independent of λ . We introduce an appropriate parametrization of the Green functions. Indicating Fourier transformation by $\sim$, e.g., for the vertex with m gluon legs

$$\begin{aligned}
 & \bar{\Gamma}_R^{(m\mathcal{A})}(x_1, \mu_1, b_1; \dots; x_m, \mu_m, b_m) \\
 &= \int_{-\pi/a}^{\pi/a} \frac{d^4 p_1}{(2\pi)^4} \dots \frac{d^4 p_m}{(2\pi)^4} (2\pi)^4 \delta_a^4 \left(\sum_{i=1}^m p_i \right) \\
 &\quad \times \exp \left[i \sum_{j=1}^m p_j \cdot (x_j + (a/2) \hat{\mu}_j) \right] \tilde{\Gamma}_R^{(m\mathcal{A})}(p_1, \mu_1, b_1; \dots; p_m, \mu_m, b_m) \quad (7.22)
 \end{aligned}$$

(δ_a represents the periodic δ -distribution), we have at the symmetric point $p^2 = q^2 = r^2$

$$\tilde{\Gamma}_R^{(3\mathcal{A})}(p, \alpha, b; q, \beta, c; r, \gamma, d) = g f^{bcd} (\delta_{\alpha\beta} (p - q)_\gamma + 2 \text{ perm}) \Lambda^{(3\mathcal{A})}(p^2) + \dots, \quad (7.23)$$

where f^{bcd} are the structure constants, $[T_b, T_c] = \sum_d f^{bcd} T_d$, and the coefficient function has an expansion in powers of $\hbar$ of the form

$$\Lambda^{(3, \mathcal{A})} = 1 + \sum_{i \geq 1} \Lambda^{(3, \mathcal{A}), i}. \quad (7.24)$$

For $i \leq n$, $\Lambda^{(3, \mathcal{A}), i}$ is finite in the large cutoff limit, and

$$\Lambda^{(3, \mathcal{A}), n+1} = r_3^{(n+1)} + \text{finite}. \quad (7.25)$$

The dots indicate contributions which carry higher powers of the momenta, their terms of the order $\hbar^{n+1}$ being finite. Similarly,

$$\begin{aligned} \tilde{\tilde{\Gamma}}_{\text{R}}^{(2, \mathcal{A})}(p, \alpha, b; -p, \beta, c) &= \delta_{bc} (-p^2 \delta_{\alpha\beta} + p_\alpha p_\beta) \Lambda^{(2, \mathcal{A})}(p^2) + \dots, \\ \tilde{\tilde{\Gamma}}_{\text{R}}^{(\mathcal{A}, N, K)}(p, \alpha, b; -p, \beta, c) &= \delta_{bc} \delta_{\alpha\beta} \Lambda^{(\mathcal{A}, N, K)}(p^2) + \dots, \\ \tilde{\tilde{\Gamma}}_{\text{R}}^{(2, \mathcal{A}, N, K)}(p, \alpha, b; q, \beta, c | r, \gamma, d) &= g f^{bcd} \{ \delta_{\alpha\beta} (p - q)_\gamma \Lambda_1^{(2, \mathcal{A}, N, K)}(p^2) \\ &\quad + (\delta_{\alpha\gamma} (p - r)_\beta - \delta_{\beta\gamma} (q - r)_\alpha) \Lambda_2^{(2, \mathcal{A}, N, K)}(p^2) \} + \dots \end{aligned} \quad (7.26)$$

$\Lambda^{(2, \mathcal{A})}$ and $\Lambda^{(\mathcal{A}, N, K)}$ have expansions similar to (7.24), whereas

$$\Lambda_i^{(2, \mathcal{A}, N, K)} = \mathcal{O}(\hbar), \quad i = 1, 2. \quad (7.27)$$

By multiplicative renormalizability we know that

$$\begin{aligned} \Lambda^{(2, \mathcal{A})} &= Z_{\mathcal{A}} \Lambda_0^{(2, \mathcal{A})}, \quad \Lambda^{(3, \mathcal{A})} = Z_{\mathcal{A}}^{3/2} \Lambda_0^{(3, \mathcal{A})} \\ \Lambda^{(\mathcal{A}, N, K)} &= (Z_{\mathcal{A}} Z_N Z_K)^{1/2} \Lambda_0^{(\mathcal{A}, N, K)} \\ \Lambda_i^{(2, \mathcal{A}, N, K)} &= Z_{\mathcal{A}} (Z_N Z_K)^{1/2} \Lambda_{0,i}^{(2, \mathcal{A}, N, K)}, \quad i = 1, 2, \end{aligned} \quad (7.28)$$

where $Z_N^{1/2}$ is the renormalization of the source N (cf. appendix B). The bare functions on the right-hand side originate from expansions for the bare vertex amplitudes similar to eqs. (7.23), (7.26). They depend on the bare parameters $\lambda_0 = Z_{\mathcal{A}}^{-1} \lambda$ and $g_0 = Z_g g$, where the renormalization constants are terminated at the order n . Using the relations (7.19) and (7.20) (where λ has to be replaced by λ_0), we get

$$\begin{aligned} 2\lambda_0 \frac{\partial}{\partial \lambda_0} \Lambda_0^{(3, \mathcal{A})} &= 3\Lambda_0^{(3, \mathcal{A})} (1 - \Lambda_0^{(\mathcal{A}, N, K)}) + p^2 \Lambda_0^{(2, \mathcal{A})} (\Lambda_{0,1}^{(2, \mathcal{A}, N, K)} + 2\Lambda_{0,2}^{(2, \mathcal{A}, N, K)}), \\ 2\lambda_0 \frac{\partial}{\partial \lambda_0} \Lambda_0^{(2, \mathcal{A})} &= 2\Lambda_0^{(2, \mathcal{A})} (1 - \Lambda_0^{(\mathcal{A}, N, K)}). \end{aligned} \quad (7.29)$$

These equations are now used to determine the λ -dependence of the contribution $r_g^{(n+1)}$ to Z_g , given by

$$2\lambda \frac{\partial}{\partial \lambda} r_g^{(n+1)} = \left[2\lambda \frac{\partial}{\partial \lambda} \log \frac{\Lambda^{(3, \mathcal{A})}}{(\Lambda^{(2, \mathcal{A})})^{3/2}} \right]_{n+1, \text{div}}, \quad (7.30)$$

where $[]_{n+1, \text{div}}$ means the divergence to the order $n+1$. To calculate eq. (7.30), we first note that

$$\frac{\partial \lambda}{\partial \lambda_0} = Z_{\mathcal{A}}^{1/2} (Z_N Z_K)^{-1/2} \frac{\Lambda^{(\mathcal{A}, N, K)}}{1 - \lambda (\partial/\partial \lambda) (\log \Lambda^{(2, \mathcal{A})})}. \quad (7.31)$$

For, using eq. (7.29) and the induction hypothesis that g_0 is independent of λ , we have

$$\frac{\partial \lambda}{\partial \lambda_0} = \frac{\partial}{\partial \lambda_0} (Z_{\mathcal{A}} \lambda_0) = \frac{\partial \lambda}{\partial \lambda_0} \lambda \frac{\partial}{\partial \lambda} \log \Lambda^{(2, \mathcal{A})} + Z_{\mathcal{A}}^{1/2} (Z_N Z_K)^{-1/2} \Lambda^{(\mathcal{A}, N, K)} \quad (7.32)$$

Thus we get

$$\begin{aligned} 2\lambda \frac{\partial}{\partial \lambda} r_g^{(n+1)} &= \left[\frac{\lambda}{\lambda_0} \frac{2\lambda_0 (\partial/\partial \lambda_0) \log [\Lambda_0^{(3, \mathcal{A})} / (\Lambda_0^{(2, \mathcal{A})})^{3/2}]}{(\partial \lambda / \partial \lambda_0)} \right]_{n+1, \text{div}} \\ &= \left[\left(1 - \lambda \frac{\partial}{\partial \lambda} \log \Lambda^{(2, \mathcal{A})} \right) \frac{\Lambda^{(2, \mathcal{A})} p^2 (\Lambda_1^{(2, \mathcal{A}, N, K)} + 2\Lambda_2^{(2, \mathcal{A}, N, K)})}{\Lambda^{(\mathcal{A}, N, K)} \Lambda^{(3, \mathcal{A})}} \right]_{n+1, \text{div}} \\ &= 0, \end{aligned} \quad (7.33)$$

because of (7.27) and the fact that $\Gamma_R^{(2, \mathcal{A}, N, K)}$ is superficially convergent. That proves our assertion by induction.

8. Matter fields

So far we have considered renormalization of only pure gauge theories. However, matter fields are easily incorporated into the discussion, and it is expected that they do not destroy renormalizability nor reveal new difficulties, as long as they are coupled in a gauge invariant manner. The part of the action which describes the matter fields should be the most general symmetric local operator of UV-degree not greater than four and IR-degree not less than four, consistent with the (lattice)

power counting. Also, renormalizability should be independent of whether the symmetry is spontaneously broken or not.

In the present section we discuss the generalizations at the example of massive fermions. For simplicity, we assume only one generation transforming under a representation of $SU(N)$. In order to apply power counting we have to avoid poles of the fermion propagators at non-zero momenta in the Brillouin zone, which is equivalent to remove the multiplicity problem of lattice fermions. This is easily done by Wilson's projection operator technique [19, 24]. By this method one adds "irrelevant" contributions to the fermionic action which remove the additional poles of naive fermions for finite lattice spacing. In the continuum limit of the renormalized theory the additional species are suppressed by infinite mass terms. We show that Wilson fermions are easily incorporated into lattice gauge theory without destroying renormalizability, i.e. there is a renormalized lattice gauge theory with Wilson fermions. As before, we get a BRS-symmetry as a consequence of the gauge symmetry of the ungauged partition function, to which an admissible gauge fixing is applied. Again the theory is perturbatively renormalized simply by adjusting the parameters of the bare action. Taking into account normalization conditions which fix the five physical parameters (finite contributions to the renormalization constants of the minimal renormalization procedure), the renormalized theory is universal in the continuum limit, which reproduces the continuum gauge theory with fermions and corresponding normalizations.

The starting point is the partition function

$$\begin{aligned}
 Z_0(J, \bar{\xi}, \xi, \bar{\phi}, \phi, K, L, \bar{P}, P; g, \lambda, m, r) \\
 = \text{const} \times \int \mathcal{D}\mathcal{A} \mathcal{D}c \mathcal{D}\bar{c} \mathcal{D}\psi \mathcal{D}\bar{\psi} \\
 \times \exp\{ \mathcal{E}_0(\mathcal{A}, c, \bar{c}, \psi, \bar{\psi}, K, L, \bar{P}, P; g, \lambda, m, r) \\
 + (1/\hbar) S_c(\mathcal{A}, c, \bar{c}, \psi, \bar{\psi}; J, \bar{\xi}, \xi, \bar{\phi}, \phi) \}, \quad (8.1)
 \end{aligned}$$

where

$$\mathcal{D}\psi \mathcal{D}\bar{\psi} = \prod_{\alpha, x} d\psi_{\alpha}(x) \cdot \prod_{\alpha, x} d\bar{\psi}_{\alpha}(x), \quad (8.2)$$

the source term S_c is as in eq. (3.4), supplemented by

$$a^4 \sum_x \text{tr}_s [\bar{\phi}(x) \psi(x) + \bar{\psi}(x) \phi(x)], \quad (8.3)$$

and $\mathcal{E}_0$ is given by (3.5) with additional contribution

$$\begin{aligned}
 & -\frac{1}{\hbar}a^4\sum_x\text{tr}_s\left\{m\bar{\psi}(x)\psi(x)+\frac{1}{2a}\sum_\mu\left[\bar{\psi}(x)(\gamma_\mu-r)D_\mu(U)\psi(x)\right.\right. \\
 & \qquad\qquad\qquad\left.\left.-\left[D_\mu(U)\psi(x)\right]^-(\gamma_\mu+r)\psi(x)\right]\right\} \\
 & +\frac{1}{\hbar}a^4\sum_x\text{tr}_s\{\bar{P}(x)s\psi(x)+s\bar{\psi}(x)P(x)\}. \tag{8.4}
 \end{aligned}$$

tr_s symbolizes the sum over all indices $\{\alpha\}$ involved of the spinors. $r \neq 0$ is the Wilson parameter and

$$\begin{aligned}
 D_\mu(U)\psi(x) &= U(x,\mu)\psi(x+a\hat{\mu})-\psi(x) \\
 [D_\mu(U)\psi(x)]^- &= \bar{\psi}(x+a\hat{\mu})U^+(x,\mu)-\bar{\psi}(x). \tag{8.5}
 \end{aligned}$$

In the $a \rightarrow 0$ -limit, the action converges to the continuum $SU(N)$ gauge field action with fermions. $\mathcal{E}_0$ is invariant under the BRS-transformation (2.21) supplemented by the following transformation laws for the fermion fields:

$$\begin{aligned}
 \delta_\varepsilon\psi(x) &= \varepsilon s\psi(x) = \varepsilon gc(x)\psi(x), \\
 \delta_\varepsilon\bar{\psi}(x) &= \varepsilon s\bar{\psi}(x) = \varepsilon g\bar{\psi}(x)c(x). \tag{8.6}
 \end{aligned}$$

That is because for $\mathcal{A}, \psi, \bar{\psi}$, eqs. (2.21) and (8.6) are infinitesimal gauge transformations with parameter $\varepsilon c(x)$. Notice the nilpotence properties

$$s^2\mathcal{A}_\mu(x) = s^2c(x) = s^2\psi(x) = s^2\bar{\psi}(x) = 0. \tag{8.7}$$

From BRS-invariance, we get the WI

$$\begin{aligned}
 \sum_x \left\{ \sum_d \left[\sum_\mu J_\mu^d(x) \frac{\partial}{\partial K_\mu^d(x)} - \bar{\xi}_d(x) \frac{\partial}{\partial L_d(x)} - \lambda F_d \left(\frac{\partial}{\partial J(x)} \right) \xi_d(x) \right] \right. \\
 \left. + \sum_\alpha \left[-\bar{\phi}_\alpha(x) \frac{\partial}{\partial \bar{P}_\alpha(x)} + \phi_\alpha(x) \frac{\partial}{\partial P_\alpha(x)} \right] \right\} Z = 0. \tag{8.8}
 \end{aligned}$$

In terms of the vertex functional $\bar{\Gamma}$, defined by

$$\bar{\Gamma} = \Gamma + a^4 \sum_{x,d} (\lambda/2) (F_d(\mathcal{A}(x)))^2 \quad (8.9)$$

and

$$\begin{aligned} W = \Gamma(\mathcal{A}, c, \bar{c}, \psi, \bar{\psi}, K, L, \bar{P}, P; g, \lambda, m, r) \\ + a^4 \sum_x \left\{ \sum_d \left[\sum_\mu J_\mu^d(x) \mathcal{A}_\mu^d(x) + \bar{\xi}_d(x) c_d(x) + \bar{c}_d(x) \xi_d(x) \right] \right. \\ \left. + \sum_\alpha [\bar{\phi}_\alpha(x) \psi_\alpha(x) + \bar{\psi}_\alpha(x) \phi_\alpha(x)] \right\}, \quad (8.10) \end{aligned}$$

where $W = \hbar \log Z$ and

$$a^4 \psi_\alpha(x) = \frac{\partial W}{\partial \bar{\phi}_\alpha(x)}, \quad a^4 \bar{\psi}_\alpha(x) = - \frac{\partial W}{\partial \phi_\alpha(x)}, \quad (8.11)$$

etc. (cf. eq. (3.17)), we get

$$\begin{aligned} \sum_x \left\{ \sum_d \left[\sum_\mu \frac{\partial \bar{\Gamma}}{\partial \mathcal{A}_\mu^d(x)} \frac{\partial \bar{\Gamma}}{\partial K_\mu^d(x)} + \frac{\partial \bar{\Gamma}}{\partial c_d(x)} \frac{\partial \bar{\Gamma}}{\partial L_d(x)} \right] \right. \\ \left. + \sum_\alpha \left[\frac{\partial \bar{\Gamma}}{\partial \psi_\alpha(x)} \frac{\partial \bar{\Gamma}}{\partial \bar{P}_\alpha(x)} + \frac{\partial \bar{\Gamma}}{\partial \bar{\psi}_\alpha(x)} \frac{\partial \bar{\Gamma}}{\partial P_\alpha(x)} \right] \right\} = 0. \quad (8.12) \end{aligned}$$

Furthermore, the Faddeev–Popov equations of motion remain unchanged. The fermion field has an UV-degree $\frac{3}{2}$ (and IR-degree 2, which implies that there are no new IR-problems due to the fermions). Thus, only the following Green functions with fermion legs are superficially divergent:

$$\begin{aligned} \langle \bar{\psi}(x) \psi(y) \rangle_{\text{1PI}}, \quad \omega = 1 \\ \langle \bar{\psi}(x) \gamma_\mu \psi(y) \mathcal{A}_\mu(z) \rangle_{\text{1PI}}, \quad \omega = 0 \\ \langle \psi(x) c(x) \rangle_{\text{1PI}}, \quad \omega = 0 \\ \langle \bar{\psi}(x) c(x) \rangle_{\text{1PI}}, \quad \omega = 0. \quad (8.13) \end{aligned}$$

(Let us mention here that the power counting theorem in the form given by ref. [8]

or [10] does not strictly apply in case of Wilson fermions because it assumes all denominators of the propagators to be of uniform lattice degree except the mass term. However, the theorem easily generalizes to the present case without change of proof, cf. appendix C.) Renormalization is described by the following

THEOREM

To every order in perturbation theory there exist renormalization constants Z_g , $Z_{\mathcal{A}}$, Z_c , Z_ψ and Z_m which except Z_m are at most logarithmically cutoff dependent, so that the renormalized action is given by

$$\begin{aligned} \mathcal{E}_R(\mathcal{A}, c, \bar{c}, \psi, \bar{\psi}, K, L, \bar{P}, P, g, \lambda, m, r) \\ = \mathcal{E}_0(Z_{\mathcal{A}}^{1/2}\mathcal{A}, Z_c^{1/2}c, Z_c^{1/2}\bar{c}, Z_\psi^{1/2}\psi, Z_\psi^{1/2}\bar{\psi}, Z_c^{1/2}K, \\ Z_\psi^{1/2}L, Z_P^{1/2}\bar{P}, Z_P^{1/2}P, Z_g g, Z_{\mathcal{A}}^{-1}\lambda, Z_m m, r), \end{aligned} \quad (8.14)$$

where $Z_P = Z_{\mathcal{A}} Z_c Z_\psi^{-1}$

As before we prove the statement in the minimal renormalization scheme. Suppose the theorem has been proved to the order $\hbar^{n-1}$, so that the WI's are satisfied. (Actually, as in the pure gauge field case we should introduce auxiliary fields of IR- and UV-degree 2, so that power counting applies. They couple to the fermions through the gauge field. However, all diagrams which have external fermion as well as auxiliary field legs are superficially convergent. We are thus allowed here to omit the auxiliary fields because we are only concerned with the modifications due to the fermions.) That constrains the counterterm to be of a form as in eq. (6.5), supplemented by

$$\begin{aligned} -a^4 \sum_{x \in aZ^4} \text{tr}_s \left\{ (r'_m + r_\psi) m \bar{\psi}(x) \psi(x) + r_\psi \frac{1}{2a} \sum_\mu (\bar{\psi}(x) \gamma_\mu \partial_\mu \psi(x) - \partial_\mu \bar{\psi}(x) \gamma_\mu \psi(x)) \right. \\ \left. + r_{\bar{\psi}\psi\mathcal{A}} \sum_\mu g \bar{\psi}(x) \gamma_\mu \mathcal{A}_\mu(x) \psi(x) + z_P g (\bar{P}(x) c(x) \psi(x) + \bar{\psi}(x) c(x) P(x)) \right\} \end{aligned} \quad (8.15)$$

The coefficients still satisfy the relations (6.7), but (8.15) implies two further equalities

$$z_1 = z_P, \quad z_2 - z_1 = r_\psi - r_{\bar{\psi}\psi\mathcal{A}}. \quad (8.16)$$

From dimensional analysis we see that all renormalization constants except r'_m are

at most logarithmically divergent. Thus we may choose by additional irrelevant contributions the counterterm to be of the form

$$\begin{aligned} \bar{\mathcal{G}}_{CT} = & \left(\frac{r_{\mathcal{A}}}{2} \frac{\partial}{\partial \tau_A} + \frac{r_c}{2} \left(\frac{\partial}{\partial \tau_c} + \frac{\partial}{\partial \tau_{\bar{c}}} \right) + \frac{r_{\psi}}{2} \left(\frac{\partial}{\partial \tau_{\psi}} + \frac{\partial}{\partial \tau_{\bar{\psi}}} \right) \right. \\ & \left. + \frac{r_K}{2} \frac{\partial}{\partial \tau_K} + \frac{r_L}{2} \frac{\partial}{\partial \tau_L} + \frac{r'_P}{2} \left(\frac{\partial}{\partial \tau_P} + \frac{\partial}{\partial \tau_{\bar{P}}} \right) + r'_m \frac{\partial}{\partial \tau_m} + r_g \frac{\partial}{\partial \tau_g} \right) \\ & \bar{\mathcal{E}}_0 \left(\tau_{\mathcal{A}} \mathcal{A}, \tau_c c, \tau_{\bar{c}} \bar{c}, \tau_{\psi} \psi, \tau_{\bar{\psi}} \bar{\psi}, \tau_K K, \tau_L L, \tau_{\bar{P}} \bar{P}, \tau_P P; \tau_g g, \tau_m m, r \right) \Big|_{\text{all } \tau=1}, \quad (8.17) \end{aligned}$$

where

$$\bar{\mathcal{E}}_0 = \mathcal{E}_0 + \hbar(\lambda/2) a^4 \sum_{x,d} (F_d(\mathcal{A}(x)))^2, \quad (8.18)$$

and the new renormalization constants are given by

$$\begin{aligned} r_L = r_{\mathcal{A}}, \quad z_2 = r_c = r_K, \quad z_1 = r_g + \frac{1}{2} r_{\mathcal{A}} + r_c, \\ z_P = \frac{1}{2} (r_c + r_{\psi} + r'_P) + r_g. \end{aligned} \quad (8.19)$$

By the induction hypothesis, the action $\mathcal{E}_{R,n-1}$ which has been renormalized to the order $n-1$ satisfies (8.14). Thus to order n

$$\begin{aligned} & \mathcal{E}_{R,n-1} + (1/\hbar) \bar{\mathcal{G}}_{CT} \\ & = \mathcal{E}_0 \left(\hat{Z}_{\mathcal{A}}^{1/2} \mathcal{A}, \hat{Z}_c^{1/2} c, \hat{Z}_{\bar{c}}^{1/2} \bar{c}, \hat{Z}_{\psi}^{1/2} \psi, \hat{Z}_{\bar{\psi}}^{1/2} \bar{\psi}, \hat{Z}_c^{1/2} K, \right. \\ & \quad \left. \hat{Z}_{\mathcal{A}}^{1/2} L, \hat{Z}_P^{1/2} \bar{P}, \hat{Z}_P^{1/2} P; \hat{Z}_g g, \hat{Z}_{\lambda} \lambda, \hat{Z}_m m, r \right) + (1/\hbar) \Delta, \quad (8.20) \end{aligned}$$

where

$$\begin{aligned} \hat{Z}_{\phi} = Z_{\phi} + r_{\phi}, \quad \phi = \mathcal{A}, c, \psi, P, g, m, \\ r_m = r'_m + O(\hbar^{n+1}), \quad r_P = r'_P + O(\hbar^{n+1}), \end{aligned} \quad (8.21)$$

and $\hat{Z}_{\lambda} = \hat{Z}_{\mathcal{A}}^{-1}$, so that $\hat{Z}_P = \hat{Z}_{\mathcal{A}} \hat{Z}_c \hat{Z}_{\psi}^{-1}$. Δ is a local function of the order $\hbar^{n+1}$, of an UV-degree not greater than four. (Note that the fermion field has an IR-degree two, which means that it causes no IR-problems. The gauge field part is handled as

before.) Hence

$$\mathcal{E}_{R,n} = \mathcal{E}_{R,n-1} + (1/\hbar) \bar{\mathcal{G}}_{CT} - (1/\hbar) \Delta$$

satisfies the theorem to the next order. That completes the proof

9. Conclusions

In the present article we have proved renormalizability of non-abelian lattice gauge theories, to every order in perturbation theory. The proof is along the lines originating from gauge invariance of the lattice cutoff. The crucial ingredients are the BRS-symmetry of the effective action derived à la Faddeev–Popov and of the functional measure, and that we have a power counting theorem at our disposal, which states how to subtract the divergencies.

The renormalizability statement is not restricted to a specific lattice action except some general constraints always to be made to get reasonable continuum limit behavior. Gauge invariance forces the gauge field action to be a function of the group elements $U(x, \mu)$ settled on the links between neighboring lattice sites. It should allow the expansion into small gauge fields. The leading order term, i.e. the naive continuum limit, is imposed to reproduce the continuum gauge field action. Then power counting is possible, not being invalidated by higher order terms. For instance, that is the case for the Wilson action, but also for even more complicated structures such as “continuum limit improved” actions ([25] and references therein), as long as the leading order term is appropriately adjusted. The divergence degrees of Feynman diagrams contributing to a particular Green function do not depend on the very details, but only on the type of that function, i.e. on the external legs, and they give rise to the same order of divergencies as in the continuum. We thus have power counting renormalizability, i.e. with increasing number of loops the order of the subtractions needed to render the large cutoff limit finite is bounded.

In weak coupling perturbation theory we are concerned with a saddle point expansion around the classical vacuum and thus restrict attention to an (infinitesimally) small neighborhood of the pure gauge orbit. Other contributions are exponentially suppressed. In order to lift the degeneracy of the saddle point we have to impose an appropriate gauge fixing. As a natural and possible gauge fixing function we have chosen the Lorentz gauge, which is linear in the gauge field, transforms under the adjoint action of the global gauge group and respects the lattice space–time symmetries. Via the Faddeev–Popov trick ghost degrees of freedom are introduced. The resulting effective action and measure exhibit invariance under BRS-transformations which for the physical fields are infinitesimal gauge transformations. Such a symmetry always exists for every gauge fixing as we should expect, because of the gauge invariance of the cutoff. All the complications of perturbative lattice gauge theory we have to live with such as the complex structure of vertices, the heap of

irrelevant diagrams and the non-trivial measure can be traced back to the nature of the cutoff and to gauge invariance of the unconstrained action. It is thus no surprise that they are under control by lattice power counting and by the WI's which follow from BRS-invariance. For instance, the contribution of the gauge field measure involves a quadratically divergent higher order "mass term" for the gauge field. This term does not only appear to make life laborious. It cancels the quadratic divergence of the gluon self-energy and implies the absence of radiative mass generation for the gluon, which would be disastrous for IR-convergence. Similar statements can be made about the other contributions. (Actually, some care is needed concerning applicability of power counting, as discussed in the text.) In general, IR-convergence can be inferred from the WI's, because they imply that vertex functions with two or three external lines, all being massless, vanish at zero external momenta. That means in the sum of all diagrams to a given order no IR-divergencies occur.

The BRS-transformation laws of the fields are non-linear. The corresponding WI written in terms of the vertex functional is a non-linear first order differential equation. Vertex amplitudes cannot be made finite by simply subtracting the divergencies without violating the WI, also if we take into account locality on the lattice. But it is always possible to add appropriate higher order terms in the coupling constant and irrelevant contributions to the action, so that a renormalized WI holds, which in the present case is identical in form to the bare one. The renormalized theory is invariant under multiplicatively renormalized BRS-transformations. From that we infer that the theory is made finite simply by renormalizing the bare parameters, independently of the originally chosen gauge field action. The conclusion is possible because all renormalization constants are at most logarithmically divergent, which insures that the terms we have to add are in fact irrelevant. Furthermore, we know from the power counting theorem that the continuum limit of the renormalized theory does not depend on the details of the lattice action. Also, the irrelevant terms do not contribute to the limit in higher orders of the loop expansion. They only have influence on the counterterms.

Clearly, universality does only hold if normalization conditions are taken into account which define the renormalized parameters. These conditions completely determine the bare parameters as a function of the renormalized ones and the cutoff (and the physical scale). Universality is also indicated by the renormalized WI's for the vertex functions. They are similar in form to the analogous identities in continuum gauge theory. Thus taking the large cutoff limit of the WI's we get the continuum ones. It is well known that the latter determine the continuum theory up to a normalization of the coupling constant and the field strengths.

The next question should be after the dependence of for instance expectation values on the gauge fixing. Knowing renormalizability of lattice gauge theories, we expect that the statements which can be made on the lattice are similar to those for other gauge invariant regularizations. For example, one may discuss gauge dependence of a priori gauge invariant operators. In general they need non-invariant

counterterms. We have discussed here only the simplest problem, namely the conventional result that in the minimal renormalization scheme the β -function does not depend on the gauge fixing parameter. Note, however, that minimal subtracted amplitudes are not universal. If normalization conditions are imposed on non-invariant Green functions, e.g. on the 3-gluon vertex to define the renormalized coupling constant g , the β -function will depend on λ and g . Nevertheless, the methods used to get such a result may probably be generalized e.g., to inquire about gauge dependence of expectation values of operators.

Renormalizability easily generalizes to the case of matter fields coupled to the gauge field in a gauge invariant way. We have proved that explicitly for Wilson fermions. They exhibit no multiplicity problem which in the present context means applicability of power counting. As before, the theory is renormalized simply by adjusting the bare parameters, where now fermion mass and field renormalization must be included. Universality of the continuum limit holds as before. In particular, the limit is independent of the Wilson parameter.

We have been concerned mainly with non-abelian gauge groups $SU(N)$ in order to deal with the problems connected with the non-linearity of the WI's. However, the method works also for abelian gauge theories coupled e.g. to Wilson fermions. Linear gauges imply that the ghost degrees of freedom decouple from the physical ones. That forces the WI for the vertex functional to be linear. Thus the representations under which the fields transform do not change under renormalization. Furthermore, the photon wave function and the abelian gauge coupling renormalization constants are related.

We finally mention that the methods presented in this article have also been applied to the two-dimensional non-linear $O(N)$ lattice σ -model [26]. We only have to take special care about IR-singularities. However, they are avoided by adding to the action a mass term for the σ -field, which easily fits into the WI's [20]. Also a very nice application of lattice power counting has been given by Marten Golterman and Don Petcher [27]. They showed that an appropriately latticised Wess–Zumino model in two dimensions becomes supersymmetric in the continuum limit to all orders of perturbation theory and without fine-tuning of the parameters of the lattice action.

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Appendix A. Absence of mass terms

Ward identities which follow from BRS-invariance are easily summarized as differential equations for the generating functionals of Green functions. That turns out to be convenient for proving renormalizability. Various relations between

different correlation functions are easily derived from them. The methods are standard, cf. e.g. refs. [17, 22, 23, 28]. Here we only show how to prove the absence of radiative mass generation for the gluon and ghost fields and that 3-point functions vanish at zero external momenta. Starting from the identities (3.18), (3.20) for the vertex functionals Γ_0 or Γ_R , by differentiating with respect to the Faddeev–Popov fields once and with respect to the gauge fields once or twice we get the identities for the desired vertex functions.

$$\sum_{\mu} \frac{1}{a} \partial_{\mu}^* \frac{\partial^2 \Gamma}{\partial K_{\mu}^d(x) \partial c_f(z)} + \frac{\partial^2 \Gamma}{\partial \bar{c}_d(z) \partial c_f(x)} = 0, \quad (\text{A.1})$$

$$\sum_{x, d, \mu} \frac{\partial^2 \Gamma}{\partial \mathcal{A}_{\nu}^e(y) \partial \mathcal{A}_{\mu}^d(x) \partial c_f(z) \partial K_{\mu}^d(x)} + a^3 \lambda \partial_{\nu}(y) \frac{\partial^2 \Gamma}{\partial c_f(z) \partial \bar{c}_e(y)} = 0, \quad (\text{A.2})$$

$$\begin{aligned} \sum_{x, d, \mu} \left[\frac{\partial^3 \Gamma}{\partial \mathcal{A}_{\rho}^h(w) \partial \mathcal{A}_{\nu}^e(y) \partial \mathcal{A}_{\mu}^d(x) \partial c_f(z) \partial K_{\mu}^d(x)} \right. \\ \left. + \frac{\partial^2 \Gamma}{\partial \mathcal{A}_{\nu}^e(y) \partial \mathcal{A}_{\mu}^d(x) \partial c_f(z) \partial K_{\mu}^d(x) \partial \mathcal{A}_{\rho}^h(w)} + \nu e y \text{ and } \rho h w \text{ interchanged} \right] \\ + a^3 \lambda \partial_{\nu}(y) \frac{\partial^3 \Gamma}{\partial c_f(z) \partial \bar{c}_e(y) \partial \mathcal{A}_{\rho}^h(w)} + \nu e y \text{ and } \rho h w \text{ interchanged} = 0. \end{aligned} \quad (\text{A.3})$$

Here Γ denotes Γ_0 or Γ_R , and $\partial_{\nu}(w)$ means lattice derivative with respect to w_{ν} . The behavior of the vertex functions for small momenta now derives from these equalities by induction on the number of loops. Namely assuming that two and three point functions vanish at zero momentum to a given order, the necessary regularity properties to the next order follow by the general discussion of ref. [11]. Inserting the corresponding small momentum expansions into the WI's immediately yields the statement.

Appendix B. Renormalization of N -insertions

We show that the operator (7.5) is multiplicatively renormalized if we omit the counterterm (7.8). That follows easily from the Faddeev–Popov equations of motion

$$\left[-\frac{1}{a} \sum_{\mu} \partial_{\mu}^* \left(\frac{\partial}{\partial K_{\mu}^d(x)} - N \frac{\partial}{\partial J_{\mu}^d(x)} \right) + a^4 \xi_d(x) - N \sum_{\mu} \frac{1}{a} \partial_{\mu}^* \frac{\partial}{\partial J_{\mu}^d(x)} \right] [1]_I = 0 \quad (\text{B.1})$$

or in terms of the vertex functional

$$F_d \left(\frac{\partial \Gamma_0}{\partial K(x)} - Na^4 \mathcal{A}(x) \right) + \frac{\partial \Gamma_0}{\partial \bar{c}_d(x)} + a^4 N F_d(\mathcal{A}(x)) = 0. \quad (\text{B } 2)$$

Assuming that all divergencies have been subtracted to the order $n-1$ in the loop expansion, the n th order counterterm with ghost lines is of the form

$$\begin{aligned} a^4 \sum_{x, d, \mu} & \left\{ K_\mu^d(x) \left(-\frac{1}{a} \left[\left(\frac{r_K + r_c + r_{\mathcal{A}}}{2} + r_g \right) \text{ad}(ag_{\mathcal{A}_\mu}(x)) + \frac{r_K + r_c}{2} \partial_\mu \right] c(x) \right)_d \right. \\ & - \bar{c}_d(x) (-1/a^2) \partial_\mu^* \left(\left[(r_c + r_g + r_{\mathcal{A}}/2) \text{ad}(ag_{\mathcal{A}_\mu}(x)) + r_c \partial_\mu \right] c(x) \right)_d \\ & \left. + N \bar{c}_d(x) \frac{1}{a} \partial_\mu^* \mathcal{A}_\mu^d(x) r_1 - N K_\mu^d(x) \mathcal{A}_\mu^d(x) r_2 \right\}. \end{aligned} \quad (\text{B.3})$$

The linear relation (B.2) implies

$$r_K = r_c, \quad r_1 = r_2 \quad (\text{B } 4)$$

and that allows us to write

$$r_1 = \frac{1}{2}(r_N + r_c + r_{\mathcal{A}}), \quad r_2 = \frac{1}{2}(r_N + r_K + r_{\mathcal{A}}), \quad (\text{B.5})$$

so that by induction, using the renormalization theorem, we can always achieve a renormalized action of the form

$$\begin{aligned} \mathcal{E}_R(\mathcal{A}, c, \bar{c}, K, L, N; g, \lambda) \\ = \mathcal{E}_0(Z_{\mathcal{A}}^{1/2} \mathcal{A}, Z_c^{1/2} c, Z_{\bar{c}}^{1/2} \bar{c}, Z_K^{1/2} K, Z_L^{1/2} L, Z_N^{1/2} N; Z_g g, Z_{\mathcal{A}}^{-1} \lambda) \end{aligned} \quad (\text{B.6})$$

(cf. (4.4)), where e.g. $Z_N = 1 + \dots + r_N + \dots$. We have not made finite the Green function

$$\left\langle \left[\sum_{x, d, \mu} \bar{c}_d(x) (1/a) \partial_\mu^* \mathcal{A}_\mu^d(x) \right] sc(y) c(z) \right\rangle_{\text{1PI}}, \quad (\text{B.7})$$

which would need a counterterm of the form (7.8). However, that has no influence on the considerations in sect. 7.

Appendix C. Generalization of the lattice power counting theorems

In ref. [8] a power counting theorem was given which assumed slightly more restrictive constraints on the form of the denominators of propagators than is actually necessary. In particular the propagators of Wilson fermions are not of the desired form. Nevertheless the theorem can easily be generalized to the following case (for a complete list of all the assumptions made by power counting theorems in the revised form cf. [29, 30]). Let C_i , $i = 1, \dots, n$ denote the denominators of the propagators with corresponding line momenta l_i . $\mu = (\mu_1, \mu_2, \dots)$ denotes a set of mass parameters and a the lattice spacing. Then the C_i are supposed to satisfy the following conditions.

$$C_i(l_i; \mu, a) = (1/a^2) \eta_i(l_i a; \mu a), \quad (\text{C.1})$$

where the η_i are smooth functions which are 2π -periodic in every component of $l_i a$ and a polynomial in μa .

$$\lim_{a \rightarrow 0} C_i(l_i; \mu, a) = l_i^2 + \mu_i^2, \quad (\text{C.2})$$

i.e. for small lattice spacing the continuum expressions are reproduced. Finally, there should exist constants $a_0 > 0$ and $K > 0$, so that

$$|C_i(l_i; \mu, a)| \geq K(\hat{l}_i^2 + \mu_i^2), \quad \mu_i^2(\mu) > 0, \quad (\text{C.3})$$

for all $a < a_0$ and all $i = 1, \dots, n$, where we have used the shorthand notation

$$\hat{k}^2 = \sum_{\mu=1}^4 \hat{k}_\mu \hat{k}_\mu, \quad \hat{k}_i = \frac{2}{a} \sin \frac{k_i a}{2}. \quad (\text{C.4})$$

The last condition looks rather technical but simply requires the denominators to be of order a^{-2} on the boundary of the Brillouin zone and that for vanishing mass μ_i the C_i has a zero only at vanishing line momentum.

With these general assumptions on the form of the denominator of the propagators the power counting theorem of ref. [8] remains valid. The same holds for the infrared power counting of ref. [10] if we allow for vanishing masses μ_i and the infrared degrees of the denominators are determined by the right-hand side of (C.3). Actually the proofs of the theorems do not change at all. Even the smoothness conditions on the denominator functions η_i may be relaxed. Namely, in case of renormalization they need to be differentiable only to such a degree that all differentiations necessary to subtract the divergencies can be done without problems.

The constraints (C.1–C.3) are satisfied by purely bosonic theories, where e.g.

$$C_i(k; \mu, a) = \hat{k}^2 + \mu_i^2, \quad (\text{C.5})$$

but also for Wilson fermions, where ($r > 0$ Wilson parameter)

$$C_i(k; \mu, a) = \sum_{\mu} \hat{k}_{\mu}^2 \cos^2 \frac{k_{\mu} a}{2} + \mu r a \hat{k}^2 + \frac{a^2 r^2}{4} (\hat{k}^2)^2 + \mu^2. \quad (\text{C.6})$$

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